Borel resummation of the long-range Ising ϵ expansion in one, two and three dimensions

Eighth-order GFF coefficients, short-range endpoint data and the critical instanton

LRI project note

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Abstract

We reconstruct the thermal scaling dimension $\Delta_{\mathcal{E}}$ of the critical long-range Ising model across its whole interacting window in $d = 1, 2, 3$, by Borel resummation of the generalized-free-field (GFF) ϵ expansion through eighth order combined with the expansion about the short-range end. In $d = 1$, with the short-range series in $\sqrt{\Delta_\sigma}$ imposed through eighth order, the conformal Borel curve gives $\Delta_{\mathcal{E}} = 0.5302$ at $\Delta_\sigma = 0.1$, with a sensitivity range $[0.5280, 0.5316]$ that is nowhere wider than 4.7×10^{-3}. This range contains every final Hamiltonian-truncation (HT) value and every converged numerical-renormalization-group (NRG) value within its quoted error, and every accepted Monte Carlo (MC) value reaches it within the sum of its statistical error and its fit-model half-range. In $d = 2$ and 3 the corresponding ranges are at most 5.2×10^{-3} and 1.5×10^{-4} wide; in $d = 2$ the way the short-range endpoint is imposed matters more, up to 7.5×10^{-3}. The critical fractional instanton supplies the candidate leading Borel singularity in each dimension; in $d = 1$ its one-loop amplitude $K_I = 6.25 \times 10^{-3}$ has the sign of every computed coefficient, which at eighth order reach 16% of its leading asymptotic form. We explain why the GFF series alone cannot reach the short-range end in $d = 1$, where $\Delta_{\mathcal{E}}$ has a square-root branch point at $\Delta_\sigma = 0$, while it nearly does in $d = 2, 3$. No HT, NRG or MC value enters a curve or is used to choose between resummation prescriptions; in $d = 1$, however, the number of imposed short-range coefficients was fixed after these data had been seen. Every quoted range is a sensitivity range, not a confidence interval.

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1 Introduction and conventions

1.1 The model and its window

We consider the ferromagnetic Ising model in d dimensions with interaction $J(r) \propto r^{-(d+s)}$. In a window $d/2 < s < s_*(d)$ the critical point is a nontrivial long-range fixed point whose spin dimension is protected,

$$x := \Delta_\sigma = \frac{d-s}{2}, \qquad \epsilon := 2s - d = d - 4x, \qquad F(\epsilon) := \Delta_{\mathcal{E}} = \sum_{n\geq 0} C_n(d)\,\epsilon^n. \tag{1.1}$$

Here $\Delta_{\mathcal{E}}$ denotes the scaling dimension of the leading $\mathbb{Z}_2$-even (thermal) operator, as in all figures, and $\nu^{-1} = d - \Delta_{\mathcal{E}}$. This operator is often called ε; we write $\mathcal{E}$ to keep it apart from the expansion parameter ϵ. At $s = d/2$ ($\epsilon = 0$) the theory is a generalized free field (GFF) and $\Delta_{\mathcal{E}} = d/2$. At the upper end s_* the long-range fixed point joins the short-range (SR) Ising universality class. Table 1 collects the windows.

d	window in s	window in $x = \Delta_\sigma$	ϵ at SR end	GFF end $(x, \Delta_{\mathcal{E}})$	SR-end data used
1	$\frac{1}{2} < s < 1$	$0 < x < \frac{1}{4}$	1	$(\frac{1}{4}, \frac{1}{2})$	$\Delta_{\mathcal{E}} = 1 + \sum_{n\leq 8} C_n^{\mathrm{SR}} x^{n/2}$
2	$1 < s < \frac{7}{4}$	$\frac{1}{8} < x < \frac{1}{2}$	$\frac{3}{2}$	$(\frac{1}{2}, 1)$	$\Delta_{\mathcal{E}} = 1 + c_2(x - \frac{1}{8})^2 + \ldots$
3	$\frac{3}{2} < s < s_*$	$S < x < \frac{3}{4}$	$3 - 4S \simeq 0.9274$	$(\frac{3}{4}, \frac{3}{2})$	$\Delta_{\mathcal{E}} = E + c_1(x - S) + \ldots$

Table 1: Interacting windows and endpoint information. In $d = 3$, $S = 0.518148806$ and $E = 1.41262528$ are the short-range Ising spin and energy dimensions and $s_* = 3 - 2S = 1.963702388$. In $d = 1$ the values at $x = 0$ are limiting values; the short-range data are a series in $\tau = \sqrt{x}$.

1.2 What resummation is for

The GFF coefficients are now known through $n = 8$ in all three dimensions [2]. They alternate in sign and grow factorially; in $d = 1$, $C_8 \simeq 8.06 \times 10^6$. The physical window, however, reaches $\epsilon = 1$, $3/2$ and 0.93 at the short-range end in $d = 1, 2, 3$. A truncated power series is useless there. Borel resummation removes the factorial growth, analytic information about the Borel transform (the location of its leading singularity, supplied by the critical instanton) extends the transform beyond its radius of convergence, and the known expansion about the short-range end pins the far side of the interval. The aim of this note is a curve $\Delta_{\mathcal{E}}(\Delta_\sigma)$ in each dimension with an honest account of how sensitive it is to the choices made, together with an understanding of which pieces of information actually control it.

The coefficients, the short-range data and the comparison values come from the literature and from the research notes of the release of Ref. [1]: the GFF, SRI, HT, NRG and MC notes [2, 3, 4, 5, 6]. This note is the resummation note of that release, and the paper's section on resummation summarizes it.

1.3 Terms used below

Approximants. CBm and PBm are the conformal Borel and Padé–Borel approximants of order m, built from $C_0, \ldots, C_m$ and any imposed endpoint data (Secs. 4.1 and 4.2); CB8 is the eighth-order conformal Borel approximant. PB$[L/M]$ is a Padé–Borel approximant with numerator degree L and denominator degree M. The Borel–Leroy parameter b_{L} (Eq. (3.1)) and the exponent α of the conformal Borel ansatz at the boundary of the map are free parameters; a scan varies them.

Endpoint constructions (Sec. 4.3) impose the short-range end. In $d = 1$: endpoint-v, resummed in $v = 1 - 2\sqrt{\Delta_\sigma}$ with the short-range series imposed through C_K^{SR}, K being the number of imposed short-range coefficients; endpoint-ϵ, the same resummed in ϵ; the archived construction

is endpoint-v with $K = 4$, used in earlier versions of this note. In $d = 2$: curvature (value, zero slope and curvature imposed), tangent (value and zero slope) and value (value only). In $d = 3$: slope (value and slope) and value. GFF-only curves resum the ϵ series directly, without short-range input.

Baseline: CB8 of endpoint-v with $K = 8$ ($d = 1$), of the curvature construction ($d = 2$) and of the slope construction ($d = 3$), at the central b_{L} and $\alpha = 0$ (Table 7).

Core band: the pointwise minimum and maximum, over orders $6, 7, 8$, three values of b_{L} and $\alpha = -1, 0, 1$, of one construction (27 approximants).

Sensitivity envelope ($d = 1$): the union of the core bands of endpoint-v for $K = 5, \ldots, 8$. Core bands and the envelope measure sensitivity, not error (Sec. 1.5).

Instanton: the critical bubble, a real stationary field of the action at negative coupling; its action A_g and the Borel-plane scale $A_\epsilon = 3A_g$ (Sec. 3).

1.4 Main results

1. **One dimension.** The baseline (CB8 of endpoint-v, with the short-range series through C_8^{SR}) gives, for example, $\Delta_{\mathcal{E}}(0.05) = 0.62223$, $\Delta_{\mathcal{E}}(0.1) = 0.53025$, $\Delta_{\mathcal{E}}(0.15) = 0.49171$, $\Delta_{\mathcal{E}}(0.2) = 0.48312$, with a shallow minimum 0.48281 at $\Delta_\sigma = 0.192$. The sensitivity envelope is 1.45×10^{-3}, 3.64×10^{-3} and 4.40×10^{-3} wide at $\Delta_\sigma = 0.05, 0.1, 0.15$ (Sec. 5.2).

2. **Two and three dimensions.** The core bands of the baselines (curvature construction in $d = 2$, slope construction in $d = 3$) are at most 5.24×10^{-3} wide (near $\Delta_\sigma = 0.308$) and 1.53×10^{-4} (near $\Delta_\sigma = 0.595$). In $d = 2$ the choice of endpoint construction moves the curve by up to 7.5×10^{-3}; for $\Delta_\sigma < 0.29$ this shift exceeds the width of the core band (Sec. 5.3).

3. **Comparisons.** In $d = 1$ the sensitivity envelope contains every final HT value and every converged NRG value within its quoted error, and every accepted MC value reaches it within the sum of its statistical error and fit-model half-range. The NRG values lie up to 7.6×10^{-4} below the baseline. At $\Delta_\sigma = 0.1$ and 0.125 they lie below the final HT values by 5.8 and 4.6 times the sum of the two quoted errors. The gap is of the size of the dependence of HT on how its truncation artifact is corrected, which the quoted HT error does not include. No comparison data favor the $K = 8$ baseline over the archived $K = 4$ construction (Table 13); $K = 8$ is motivated a priori only, and it was fixed after these data had been seen. In $d = 2$ finite-cutoff HT lies above every endpoint construction at $s = 1.6$–1.65, by about 3×10^{-3} beyond the highest of their core bands; its cutoff dependence there does not show that the excess survives at infinite cutoff. Published MC values (quoted errors 0.006 to 0.034) cannot separate the curves. In $d = 3$ the project's MC point at $s = 1.6$ lies 1.0×10^{-3} above the curve, and bootstrap kinks read off a figure lie 0.004 to 0.010 below it at $s = 1.59$–1.87, the gap growing toward the short-range end (Sec. 6).

4. **Instanton.** The critical real bubble gives the candidate Borel singularity $-A_\epsilon$ with $A_\epsilon = 9\pi^2/\Gamma(\frac{1}{4})^4$, $\frac{9}{8}$ and $\frac{9}{4}[\Gamma(\frac{9}{4})/\Gamma(\frac{3}{4})]^2$ for $d = 1, 2, 3$. In $d = 1$ the regular one-instanton sector predicts $C_n \sim K_I(-1/A_\epsilon)^n\Gamma(n + 4)$ with $K_I = 6.2466 \times 10^{-3}$; the coefficients have the predicted sign, but at $n = 8$ they are only 0.158 of this leading form and still rising (Sec. 3).

5. **Analytic structure.** In $d = 1$, $\Delta_{\mathcal{E}}$ has a square-root branch point at $\Delta_\sigma = 0$; without short-range input the GFF series cannot reproduce it and the GFF-only curve ends at 0.567 instead of 1. In $d = 2, 3$ the short-range end is approached analytically and the GFF-only curve reaches 1.0049 (exact 1) and 1.4161 (exact 1.4126) (Sec. 4.4).

1.5 Status of the statements

Table 2 separates what is established from what is conditional and from what is our interpretation. We use the following words throughout: a *band*, *envelope* or *range* is the pointwise minimum and maximum over a stated set of resummation prescriptions. It measures sensitivity to those choices; it is not a confidence interval, it has no probabilistic coverage, and it does not bound the error of the approximant. Printed digits identify an approximant reproducibly and are not claimed digits of the physical dimension. No HT, NRG or MC value enters any curve or is used to choose between prescriptions. In $d = 1$, however, the number $K = 8$ of imposed short-range coefficients was fixed after these data had been seen; it is motivated a priori only (Secs. 5.2 and 6.2). Uncertainties of different kinds are never added in quadrature: a difference is compared with each quoted error separately or with their linear sum, as stated each time (Sec. 6).

Statement	Status
Critical bubble and its action A_g	Explicit classical solution; least action among nonzero real finite-energy stationary fields (sharp fractional Sobolev inequality).
$A_\epsilon = 3A_g$ as the nearest singularity of the full Borel transform	Conditional. It is the exponential scale of the one-instanton sector; dominance is not proved.
$b_I = (d + 5)/2$	$d = 1$: formal renormalized one-instanton prediction, with the leading mass insertion checked. $d = 2, 3$: power counting only.
$K_I = 6.25 \times 10^{-3}$ $(d = 1)$	One-loop prediction of the stated prescription, which reproduces the ϕ^4 amplitude in $d = 4$; the treatment of the regulator dependence of the bubble action (Sec. 3.4) is argued, not derived in full. No $d = 2, 3$ amplitude.
Square-root branch point at $\Delta_\sigma = 0$ $(d = 1)$	Established within the short-range construction: odd powers of $\sqrt{\Delta_\sigma}$ come from an eigenvalue splitting.
Fixed-point merger of Kosterlitz–Thouless type at $s = 1$	Our interpretation of that structure.
Radius $\simeq \frac{1}{2}$ of the short-range series in $\sqrt{\Delta_\sigma}$	Suggestive only (eight coefficients).
$K = 8$ in $d = 1$	Motivated a priori (all available short-range coefficients); fixed after the comparison data had been seen.
Eighth-order curves	Reproducible approximants; not fitted to comparison data.
Bands and envelopes	Sensitivity ranges, not confidence intervals.
C_9	Not computed; all C_9 values are predictions or scenarios.

Table 2: Logical status of the main statements.

Earlier versions. This note merges and supersedes three earlier project notes, for $d = 1$ (7 October 2026) and for $d = 2$ and $d = 3$ (12 October). Four things have changed since then. The short-range series in $d = 1$ is imposed through C_8^{SR} ($K = 8$) instead of C_4^{SR} (the archived $K = 4$ construction, kept as a control). $C_8(1)$ is the final value of Sec. 2 instead of a four-batch value. The $d = 1$ HT and NRG values are the final ones of the HT and NRG notes [4, 5], which replace earlier extrapolations. And the statement that $K = 8$ removes an offset of $K = 4$ against HT and NRG is withdrawn (Sec. 6.2).

2 Inputs

2.1 GFF coefficients

Table 3 lists the coefficients used. In every dimension $C_0 = d/2$ and $C_1 = -1/6$. The coefficients C_0 to C_3 are published [7]; C_4 to C_8 are from the GFF note [2]. Through C_5 they are exact expressions in an explicit master basis; the lowest ones read

$$C_2(d) = \frac{\mathcal{A}_d}{9}, \qquad C_3(d) = \frac{\mathcal{A}_d^2 + \mathcal{R}_d - 8Q_d}{108}, \qquad \mathcal{A}_d = \psi(\tfrac{d}{2}) + \psi(1) - 2\psi(\tfrac{d}{4}), \quad \mathcal{R}_d = \psi_1(\tfrac{d}{2}) - \psi_1(1), \tag{2.1}$$

with Q_d the three-loop primitive period (one third of the tetrahedron period, as corrected in the erratum of Ref. [7]). For $d = 1$ this gives $\mathcal{A}_1 = \pi + 4\log 2$; for $d = 2$, $C_2 = \frac{4}{9}\log 2$. The values of C_5 are numerical evaluations of the exact expression ($d = 3$: a supplied evaluation of the master basis). The coefficients C_6, C_7, C_8 are numerical estimators with empirical allowances.

n	$C_n(1)$	$C_n(2)$	$C_n(3)$
0	1/2	1	3/2
1	−1/6	−1/6	−1/6
2	0.6571312639810638	0.3080654135821979	0.181221785405554
3	−3.719856322901833	−0.8748500126526337	−0.353581520433173
4	45.42488010356	5.048868431354847	1.22806354532762
5	−731.13947662454	−38.83364589622536	−5.85035142075133
6	$14331.68703276 \pm 4.9 \times 10^{-7}$	362.36647 ± 0.004	33.57543 ± 0.0005
7	-322549.889 ± 41	-3876.126 ± 2.1	-220.3545 ± 0.13
8	8059766.7 ± 17000	45990.0 ± 33	1602.605 ± 3.9

Table 3: GFF coefficients used [2]. The ± entries are empirical reporting allowances, not standard deviations: for the Monte Carlo estimates ($C_6(2)$, $C_6(3)$, C_7, C_8) four sampling standard errors, rounded up; for the deterministic graph sum $C_6(1)$ a convergence-and-rounding allowance. Sampling standard errors: $C_6(2)$ 8.0×10^{-4}; $C_6(3)$ 1.19×10^{-4}; $C_7(1)$ 10.09; $C_7(2)$ 0.524; $C_7(3)$ 0.0318; $C_8(1)$ 4210; $C_8(2)$ 8.22; $C_8(3)$ 0.952. Full stored digits are used in the calculations.

Three remarks on Table 3. First, $C_6(1)$ is a deterministic sum over all six-loop graphs with a convergence-and-rounding allowance; a closed form in nine special-function values gives 14331.68703275805, which differs by -1.03×10^{-8} and is not used. Second, the final value $C_8(1) = 8059766.723129955$ is the mean of fourteen production batches (sampling standard error 4209.6, pre-registered allowance ±17000, quoted $(8.060 \pm 0.017) \times 10^6$). It replaces the four-batch value 8066723.343 ± 33000 used in earlier versions of this note; every one-dimensional result below uses the final value. Third, the eighth-order Monte Carlo campaigns, whose C_8 includes the power-containing graph sector of the GFF note, also estimate the lower coefficients; those estimates serve as cross-checks only, and the more precise dedicated values of $C_0, \ldots, C_7$ in the table are used. No joint covariance of C_6, C_7, C_8 is available, so coefficient sensitivities below are sums of absolute responses to the allowances.

2.2 Short-range data

One dimension. Near $s = 1$ the thermal dimension is a series in $\tau = \sqrt{x}$ [3],

$$\Delta_{\mathcal{E}} = 1 + \sum_{n \geq 1} C_n^{\mathrm{SR}}\, \tau^n, \qquad \begin{aligned} &C_1^{\mathrm{SR}} = -2, \quad C_2^{\mathrm{SR}} = \tfrac{1}{2}, \quad C_3^{\mathrm{SR}} = \tfrac{39}{16} + 2\zeta(2), \\ &C_4^{\mathrm{SR}} = 4 - 4\zeta(2) - 8\zeta(3), \end{aligned} \tag{2.2}$$

with closed forms in multiple zeta values through C_7^{SR}. Numerically $C_{3,\ldots,8}^{\mathrm{SR}} = 5.72736813$, -12.19619149, 29.23781158, -58.66569747, 51.06898861 and 247.38120480. The leading orders

reproduce the published dual description [8, 9], $\Delta_\pm = 1 \pm \sqrt{2(1-s)} + (1-s)/4 + O((1-s)^{3/2})$, with $1 - s = 2x$. The higher coefficients are results of the project's construction and are conditional on its regulator, matching and two-coupling renormalization assumptions, which the SRI note states in its Appendix A and examines in its Appendix D, "The two-coupling closure and the prescription" [3]. C_8^{SR} is a completed numerical assembly; a short zeta-value form agrees with it to a recorded residual of about 10^{-86} but is unproved. An independent re-derivation gives $C_8^{\mathrm{SR}} = 242.8 \pm 2.4$ (one statistical standard error), 1.9 standard errors lower [3]; using its central value instead moves the baseline curve by at most 3×10^{-5}, so this tension does not matter here.

The magnitudes $|C_n^{\mathrm{SR}}|^{1/n}$ are 1.96, 1.97, 1.75 and 1.99 for $n = 5, 6, 7, 8$. This suggests a radius of convergence close to $\frac{1}{2}$ in τ, that is, singularities at $\tau = \pm\frac{1}{2}$: $\tau = \frac{1}{2}$ is the GFF point $\Delta_\sigma = \frac{1}{4}$, and $\tau = -\frac{1}{2}$ is its image on the second sheet (Sec. 4.4). With eight coefficients this is suggestive, not a determination.

Two dimensions. With $\delta = x - \frac{1}{8}$ [10, 3],

$$\Delta_{\mathcal{E}} = 1 + c_2\,\delta^2 + O(\delta^3), \qquad c_2 = -4.066270148132521 55274\ldots, \tag{2.3}$$

with zero linear term, so $F(\frac{3}{2}) = 1$, $F'(\frac{3}{2}) = 0$ and $F''(\frac{3}{2}) = c_2/8$ in ϵ derivatives. The vanishing linear term is published [10]; the value of c_2, from the SRI note [3], is a formal result in a critically tuned analytic minimal-subtraction prescription. A Monte Carlo evaluation of the regulated integral gives $c_2 = -4.075 \pm 0.019$ (one statistical standard error) [3]. A uniform proof of the nested subtractions is not available.

Three dimensions. With $S = 0.518148806(24)$ and $E = 1.41262528(29)$ from the conformal bootstrap [11] and $\delta = x - S$,

$$\Delta_{\mathcal{E}} = E + c_1\,\delta + O(\delta^2), \qquad c_1 = \frac{a_0}{b_0} = 0.27315242382601645, \tag{2.4}$$

where $a_0 = 3.3474594916917857$ and $b_0 = 12.254914105481044$ are the crossover coefficients of $\gamma_{\mathcal{E}} = a_0 g_{\mathrm{SR}}^2 + \ldots$ and $\beta_{\mathrm{SR}} = -\delta g_{\mathrm{SR}} + b_0 g_{\mathrm{SR}}^3 + \ldots$ in the scaling theory of Ref. [10], computed in the SRI note [3] from the 3d Ising spectrum of Refs. [12, 11]. That note quotes a working range $c_1 \in [0.2690, 0.2760]$, conditional on its spectral and truncation assumptions (60 retained primaries through dimension 24, gaps for omitted operators, inputs within twice their errors); it is not a statistical interval.

3 Large orders and the critical instanton

3.1 Borel transform and the dispersion relation

A large-order behavior $C_n \sim \kappa(-1/A)^n \Gamma(n + b + 1)$ corresponds to a singularity of the Borel transform at $t = -A$. With the Borel–Leroy transform

$$\mathcal{B}_{b_{\mathrm{L}}}(t) = \sum_{n \geq 0} \frac{C_n\, t^n}{\Gamma(n + b_{\mathrm{L}} + 1)}, \qquad \mathcal{R}[F](\epsilon) = \int_0^\infty e^{-u} u^{b_{\mathrm{L}}}\, \mathcal{B}_{b_{\mathrm{L}}}(\epsilon u)\, \mathrm{d}u, \qquad b_{\mathrm{L}} > -1, \tag{3.1}$$

termwise integration returns $C_n \epsilon^n$. If the discontinuity of F across the negative axis has a sector $\mathrm{Disc}\, F(-u) \sim D\, u^{-p} e^{-A/u}$, a dispersion relation gives the contribution

$$C_n^{(I)} \sim (-1)^{n+1}\, \frac{D}{2\pi i}\, A^{-n-p}\, \Gamma(n + p), \tag{3.2}$$

so $b = p - 1$. The instanton supplies the candidate location $-A$ and, after a fluctuation and renormalization analysis, a candidate strength. It constrains the tail of the series; it does not supply the missing coefficients. Eight coefficients cannot determine the singularity by themselves: adding $\eta[(1 + t/t_0)^{-4} - T_8(1 + t/t_0)^{-4}]$, with T_8 the Taylor truncation, to the Borel transform reproduces all known terms and places a singularity at $t = -t_0$ for any $t_0 > 0$.

3.2 The critical bubble in d dimensions

At the expansion point $s = d/2$ normalize the free covariance to $G(x) = |x|^{-d/2}$, so that the kinetic operator is $\mathcal{K} = |D|^{d/2}/(2\pi)^{d/2}$, where $|D|^\rho$ has symbol $|p|^\rho$. At negative quartic coupling the action $\mathcal{S} = \frac{1}{2}\int \phi\mathcal{K}\phi - \frac{h_{\rm phys}}{4!}\int \phi^4$ has the stationary family

$$\phi_I(y) = \pm\frac{B_d}{\sqrt{h_{\rm phys}}}\left[\frac{R}{R^2 + |y - y_0|^2}\right]^{d/4}, \qquad B_d^2 = \frac{6\,\Gamma(3d/4)}{\pi^{d/2}\,\Gamma(d/4)}, \tag{3.3}$$

labeled by position, size and sign. In the project coupling g, normalized so that $\beta_g = -\epsilon g + 3g^2 + \ldots$ and $g_* = \epsilon/3 + \ldots$, the action at $g = -h$ is $A_g(d)/h$ with

$$A_g(d) = \frac{3}{2}\,\frac{\Gamma(3d/4)^2}{\Gamma(d/4)^2\,\Gamma(d)}, \qquad A_\epsilon = 3A_g. \tag{3.4}$$

The candidate Borel singularity of the physical ϵ series is at $t = -A_\epsilon$:

$$\begin{aligned} A_\epsilon(1) &= \frac{9\pi^2}{\Gamma(\frac{1}{4})^4} = 0.51406490367502158, \qquad A_\epsilon(2) = \frac{9}{8}, \\ A_\epsilon(3) &= \frac{9}{4}\left[\frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{3}{4})}\right]^2 = 1.92343325557014548, \end{aligned} \tag{3.5}$$

and $A_\epsilon(4) = 3$ in the same normalization. In $d = 1$, $A_g = 3\pi^2/\Gamma(\frac{1}{4})^4 = 0.171354967891674$; the normalization was checked independently by rescaling the field and the coupling.

Two properties fix the meaning of this saddle. The sharp fractional Sobolev (Hardy–Littlewood–Sobolev) inequality $T \geq \kappa_{\rm S}\sqrt{V}$, with $T = \int \phi\mathcal{K}\phi$ and $V = \int \phi^4$, combined with the virial identity $T = hV/6$ of any nonzero real stationary field, gives $\mathcal{S} \geq A_g/h$ with equality on the bubble [13, 14]. The bubble is therefore the least-action nonzero real stationary field in the finite-energy class. This does not classify complex saddles, integration cycles or the singularities of the renormalized theory. Second, dilations give $(2s - d)T = 0$ for a finite-action massless solution, so the closed form exists only at $s = d/2$. Using it for the large-order behavior of the expansion *about* $\epsilon = 0$ is consistent, but continuing the saddle to physical s requires a regulated (constrained) treatment.

3.3 Fluctuations, power counting and the exponent b_I

Mapped conformally to the sphere, the Hessian relative to the free kinetic form has eigenvalues $1 - 3p_0/p_\ell$ with $p_\ell = \Gamma(\ell + 3d/4)/\Gamma(\ell + d/4)$: one negative mode ($\ell = 0$), $d+1$ zero modes ($\ell = 1$: d translations and the dilation) and positive modes for $\ell \geq 2$. Bubble nondegeneracy excludes further zero modes in its class [15]. The $d+1$ collective coordinates, normalized with the kinetic metric (the flat L^2 norm of the dilation mode diverges), contribute $h^{-(d+1)/2}$; the four amputated legs contribute h^{-2}. After the size integral and the minimal-subtraction residue, $\mathrm{Disc}\,\beta(-h) \propto h^{-(d+5)/2}e^{-A_g/h}$. The fixed-point shift divides by $\beta'(g_*) = O(\epsilon)$ and adds one inverse power, so

$$\mathrm{Disc}\,F(-u) \propto u^{-(d+7)/2}e^{-A_\epsilon/u}, \qquad C_n^{(I)} \sim K_I\,(-1/A_\epsilon)^n\,\Gamma(n + b_I + 1), \qquad b_I = \frac{d+5}{2}. \tag{3.6}$$

This gives $b_I = 3, \frac{7}{2}, 4$ for $d = 1, 2, 3$ and reproduces the established $\frac{9}{2}$ of the short-range ϵ expansion in $d = 4$ [16, 18].

In $d = 1$ the analysis goes beyond counting. It includes the kinetic-metric measure $G_{ij} = \frac{3A_g}{8hR^2}\delta_{ij}$, the cancellation of the determinant pole $3A_g/z$ against the one-loop coupling counterterm (an independent check of the one-loop coefficient 3), and the leading direct mass insertion, whose discontinuity is down by one power of u and therefore cannot cancel the leading term within the same regular semiclassical expansion. Three different statements must be kept apart:

1. $b_{\mathrm{L}} = 3$ is a *choice* of the freely adjustable Borel–Leroy parameter;
2. $b_I = 3$ is the exact integer predicted by the regular one-instanton sector, not a fit;
3. that this sector controls the complete coefficient sequence has not been proved. It requires a uniform renormalized saddle expansion and global dominance with nonzero net amplitude; competing complex saddles, multi-instanton sectors and non-instanton mechanisms have not been classified [19].

In $d = 2, 3$ only the action and the power counting are available; no renormalized determinant, Stokes amplitude or competing-singularity analysis has been done.

3.4 The one-loop amplitude in one dimension

Following the minimal-subtraction procedure of Refs. [16, 17] with the project's analytic regulator $G_z(y) = |y|^{-(1-z)/2}$, the leading one-instanton sector gives

$$\begin{aligned} C_n &\sim K_I\,(-1/A_\epsilon)^n\,\Gamma(n+4), \qquad K_I = 6.2466 \times 10^{-3}, \\ \beta_n &\sim (-1)^{n+1} C_\beta\, A_g^{-(n+1)} \Gamma(n+4), \qquad C_\beta = 0.024373, \end{aligned} \tag{3.7}$$

the second relation for the coefficients β_n of the minimal-subtraction beta function $\beta_0(g) = \sum_n \beta_n g^{n+1}$. The assembly is

$$K_I = \frac{\mathcal{N}\, e^{-2\mathcal{A}_1 A_g}}{3\pi A_g}, \qquad C_\beta = \frac{\mathcal{N}}{\pi}, \qquad \mathcal{N} = \frac{9}{2\sqrt{2}\,\pi^2 A_g} \exp\Big(-\tfrac{1}{2}L_{\mathrm{sub}} + 9A_g \log 2 - \tfrac{21}{4} + \delta_{\mathrm{p}}\Big) = 0.076571, \tag{3.8}$$

with the following ingredients:

- $L_{\mathrm{sub}} = -1.769169774220360678$, the convergent subtracted spectral sum over the positive modes; its terms fall like $n^{-3/2}$, so it is summed with an exact asymptotic tail (a default automatic summation gets the third decimal wrong);
- the finite part $9A_g \log 2 = 1.06897$ of the regulated bubble trace, after its pole $3A_g/z$ cancels against the coupling counterterm, and the subtraction $\Sigma_{\mathrm{spec}} = \frac{21}{2}$ of the modes $n = 0, \pm 1$, which enters the exponent as $-\Sigma_{\mathrm{spec}}/2 = -\frac{21}{4}$;
- the prefactor $9/(2\sqrt{2}\pi^2 A_g) = 2 \cdot \frac{1}{2\sqrt{2}} \cdot \frac{3}{16\pi} \cdot \frac{24}{\pi} \cdot \frac{1}{A_g}$: the negative mode, $1/(2\sqrt{2})$ in magnitude per saddle sign, with both signs contributing equally; the collective-coordinate measure $\sqrt{\det G}/(2\pi) = 3A_g/(16\pi h R^2)$; and the four amputated legs, $(24/\pi A_g)(A_g/h)^2 R$. The remaining factor $(A_g/h)^3$ enters the size integral, whose simple pole leaves $(A_g/h)^2 e^{-A_g/h}$;
- the constrained-profile shift $\delta_{\mathrm{p}} = \frac{3}{2}A_g(\pi - 4\log 2) = 0.0948460$;
- the fixed-point factor $e^{-2\mathcal{A}_1 A_g} = 0.13175005$, with $\mathcal{A}_1 = \pi + 4\log 2$.

The positive K_I fixes the sign of C_n to that of $(-1)^n$, in agreement with all computed coefficients.

The same programs, with only dimension-specific inputs changed, reproduce three known results: the zero-dimensional quartic integral; McKane's general ϕ^6 formula in $d = 3$ to fifteen digits; and the minimal-subtraction beta-function amplitude of ϕ^4 theory in $d = 4 - \epsilon$ to twelve digits, once the corresponding constrained-profile shift ($-\frac{9}{2}$ there) is included [16, 17]. The remaining qualification concerns the regulator dependence of the classical action, $A(z) = A_g + z\vartheta A_g$ with $\vartheta = \frac{\pi}{2} - 2\log 2$. The adopted prescription treats it as a rescaling of the size coordinate, which drops out of the simple-pole residue; this cancellation is argued and supported by the $d = 4$ benchmark, but not yet derived while tracking the integration limits, determinant, measure, legs and counterterms explicitly. The variants $K_I = 5.681 \times 10^{-3}$ (without δ_{p}) and 5.167×10^{-3} (with $-\delta_{\mathrm{p}}$ in place of δ_{p}) are disfavored by the adopted derivation, not excluded; their difference is not an error bar. No $d = 2$ or $d = 3$ amplitude has been computed, and the one-dimensional value is not borrowed.

3.5 Finite-order mismatch

Table 4 compares the one-dimensional coefficients with the leading form. Every coefficient has the predicted sign. In magnitude they reach 10% to 16% of $K_I(-1/A_\epsilon)^n\Gamma(n+4)$, rising from $n = 4$ without a plateau. The column with the alternative normalization $n!\,n^3$ is not an independent test: the two denominators differ by $(n+1)(n+2)(n+3)/n^3$, which is 1.93359375 at $n = 8$. For the minimal-subtraction beta function of ϕ^4 theory in $d = 4$ the corresponding fractions at one to six loops are 43%, 12.5%, 9.6%, 9.4%, 10.4% and 12.1% [18]. The similarity makes a slow approach to the asymptotic regime plausible; it does not show that both theories approach it in the same way.

n	$\dfrac{C_n}{K_I(-1/A_\epsilon)^n\Gamma(n+4)}$	$\dfrac{C_n}{K_I(-1/A_\epsilon)^n\,n!\,n^3}$	$r_n = -\dfrac{C_n}{nC_{n-1}}$	$\widehat{a}_n = -\dfrac{C_n}{(n+3)C_{n-1}}$
3	0.112	0.499	1.887	0.943
4	0.101	0.331	3.053	1.744
5	0.104	0.280	3.219	2.012
6	0.117	0.272	3.267	2.178
7	0.135	0.283	3.215	2.251
8	0.158	0.305	3.123	2.272

Table 4: One-dimensional coefficients against the leading instanton form, with the final $C_8(1)$. If Eq. (3.6) already dominated, $\widehat{a}_n$ would equal $1/A_\epsilon = 1.9453$.

The power-corrected ratio $\widehat{a}_8$ lies 16.8% above $1/A_\epsilon$ and is still increasing. Write the next correction as $C_n \sim K_I(-1/A_\epsilon)^n\Gamma(n+4)[1 + d_1A_\epsilon/(n+3) + \dots]$. It changes $\widehat{a}_nA_\epsilon$ by $-d_1A_\epsilon/[(n+2)(n+3)]$, so if it alone explained the drift, $n^2(\widehat{a}_nA_\epsilon - 1)$ would approach the constant $-d_1A_\epsilon$; it takes the values 4.31, 7.69 and 10.74 at $n = 6, 7, 8$. The coefficient d_1 is only partly known: its known pieces (fixed-point composition and the residue operation) sum to -6.40, a -30% correction at $n = 8$; the two-loop fluctuations, the subleading size integral and the direct mass insertion are not computed. Fits of the first column of Table 4 with corrections in powers of $1/(n+3)$ are unstable, depending on which orders are fitted. One correction term gives an asymptotic value 0.24 to 0.34 (and $d_1 \simeq -8.5$ to -11.6); two terms give 0.58 to 0.73 (and $d_1 \simeq -22$ to -25), values of d_1 that would give corrections beyond -100% at $n = 8$. The eighth order is therefore not yet a sharp test of K_I.

The same diagnostic in $d = 2$ and 3, with $\widehat{a}_n = -C_n/[(n+b_I)C_{n-1}]$ (in $d = 1$, $b_I = 3$) and the candidate b_I, gives Table 5. At $n = 8$ the ratio exceeds the candidate limit by 16.1% in $d = 2$ and 16.6% in $d = 3$, close to the 16.8% of $d = 1$, and is still rising in all three dimensions. Eighth order does not establish asymptotic dominance in any dimension, and no amplitude is inferred from these ratios.

n	2	3	4	5	6	7	8
$d = 2$, $\widehat{a}_n$ (limit 0.8889)	0.3361	0.4369	0.7695	0.9049	0.9822	1.0187	1.0317
$d = 3$, $\widehat{a}_n$ (limit 0.5199)	0.1812	0.2787	0.4342	0.5293	0.5739	0.5966	0.6061

Table 5: Corrected ratios $\widehat{a}_n = -C_n/[(n+b_I)C_{n-1}]$ in $d = 2$ ($b_I = \frac{7}{2}$) and $d = 3$ ($b_I = 4$), against the candidate limit $1/A_\epsilon$.

4 Resummation methods

4.1 Padé–Borel and admissibility

For the series $f(z) = \sum_{n\le m} f_n z^n$ actually being resummed, the Padé–Borel (PB) approximant replaces $\mathcal{B}_{b_L}$ of Eq. (3.1) by a rational function $[L/M]$ with $L + M = m$. It receives no instanton information. Candidates have denominator degree $M = \lfloor m/2\rfloor - 1, \ldots, \lfloor m/2\rfloor + 1$ (one more for odd m), restricted to $1 \le M \le m$, ordered by $(|m - 2M|, M)$; the first candidate without a positive-real denominator root, defined by $\Re t > 0$ and $|\Im t| < 10^{-8}\max(1, |t|)$, is selected. No principal value is taken. If no candidate survives, the configuration is reported as unavailable, never replaced silently. All poles and rejected candidates are archived.

PBm denotes the approximant this rule selects at order m. The rule fixes the PB controls of the curves in $d = 2, 3$ (PB8 is $[5/3]$ in $d = 2$ at $b_L = 3.5$; PB7 is $[3/4]$ in $d = 3$ at $b_L = 4$) and the shapes of all held-out predictions (Sec. 7). In $d = 1$ the curve controls have fixed shapes: those the rule selects for the archived $K = 4$ construction at $b_L = 3$, $[4/2]$, $[5/2]$ and $[5/3]$ at orders 6, 7 and 8, held fixed for every K so that a matched comparison changes only the Borel approximation. At order 8 the rule itself selects $[5/3]$ also for $K = 5$–7; at $K = 8$ only $[3/5]$ survives.

4.2 Conformal Borel

Assuming that the Borel transform is analytic in the plane cut along $(-\infty, -A]$, the map

$$w(t) = \frac{\sqrt{1 + t/A} - 1}{\sqrt{1 + t/A} + 1}, \qquad t = \frac{4Aw}{(1 - w)^2}, \tag{4.1}$$

sends the cut to the unit circle and the positive integration axis into the disk. The conformal Borel (CB) approximant is the Taylor-matched polynomial $\widehat{\mathcal{B}}(t) = (1 - w)^{-\alpha} \sum_{k\le m} p_k w^k$ (Appendix A), integrated as in Eq. (3.1). It uses the candidate action A, but the amplitude K_I is not imposed. The baseline has $\alpha = 0$; α parametrizes the behavior at $w = 1$ and is not a derived exponent. A CB versus PB comparison compares two approximation families, not the effect of one isolated input. The map can fail if the assumed nearest singularity is wrong, if there are other off-axis singularities, or if subleading terms are large.

4.3 Endpoint constructions

All constructions below preserve every supplied GFF coefficient through order m: the series of the remainder is formed by exact series operations, truncated after each step. Regularity of the resummed remainder at the short-range end then enforces the endpoint constraints. Imposing endpoint information is supplied input; agreement at that endpoint is not a prediction.

One dimension. With $F_K^{\mathrm{SR}}(x) = \sum_{n\le K} C_n^{\mathrm{SR}}\, x^{n/2}$ (and $C_0^{\mathrm{SR}} = 1$),

$$\begin{gathered} H = \frac{F - F_K^{\mathrm{SR}}}{(4x)^{(K+1)/2}}, \qquad v = 1 - \sqrt{1 - \epsilon} = 1 - 2\sqrt{x}, \quad \epsilon = 2v - v^2, \\ F = F_K^{\mathrm{SR}} + (4x)^{(K+1)/2}\, \mathcal{R}_m\big[H(2v - v^2)\big](v). \end{gathered} \tag{4.2}$$

The series of H in ϵ is composed with $\epsilon = 2v - v^2$ and resummed in v. The action transforms with the expansion variable: since $A_\epsilon/\epsilon(v) = A_\epsilon/(2v) + A_\epsilon/4 + O(v)$, the CB map in v uses $A_v = A_\epsilon/2 = 0.2570324518375$. This change of the physical expansion variable is distinct from the conformal map of the Borel plane. The archived construction of earlier versions of this note is $K = 4$, with $F_4^{\mathrm{SR}} = 1 - \sqrt{q} + \frac{q}{8} + (\frac{39}{128} + \frac{\pi^2}{24})q^{3/2} + (\frac{1}{4} - \frac{\pi^2}{24} - \frac{\zeta(3)}{2})q^2$, $q = 4x$. The baseline of this note is $K = 8$, which uses all available short-range coefficients. A control resums H in ϵ instead of v (endpoint-ϵ), and the GFF-only control resums F directly in ϵ.

Two dimensions. With $q = 1 - \epsilon/\frac{3}{2} = (x - \frac{1}{8})/\frac{3}{8}$ and $F^{\rm SR} = 1 + \frac{9c_2}{64}q^2$, the baseline (*curvature*) construction is $F = F^{\rm SR} + q^3\mathcal{R}_m[(F - F^{\rm SR})/q^3]$; it imposes value, zero slope and curvature. The *tangent* construction $F = 1+q^2\mathcal{R}_m[(F-1)/q^2]$ imposes value and zero slope; the *value* construction $F = 1 + q\,\mathcal{R}_m[(F-1)/q]$ imposes only the value; the GFF-only curve resums F itself. All resum in ϵ with $A_\epsilon = \frac{9}{8}$.

Three dimensions. With $X = \frac{3}{4}-S$, $q = (x-S)/X = 1-\epsilon/\epsilon_*$ ($\epsilon_* = 3-4S$) and $F^{\rm SR} = E+c_1Xq$, the baseline (*slope*) construction is $F = F^{\rm SR} + q^2\mathcal{R}_m[(F - F^{\rm SR})/q^2]$. The *value* construction is $F = E + q\,\mathcal{R}_m[(F - E)/q]$, and the GFF-only curve resums F. All resum in ϵ with $A_\epsilon(3)$. In $d = 2, 3$ the one-dimensional square-root variable is not used, for the reason explained next.

4.4 Analytic structure at the short-range end

What is established in $d = 1$. In the project's construction of the short-range end, the two lowest even operators are governed by a two-coupling flow whose stability matrix at the fixed point has trace $T = (1-s)/2+O((1-s)^2)$ and determinant $D = -2(1-s)+O((1-s)^2)$, with $1-s = 2x$. The fixed-point fugacity is of order $\sqrt{1-s}$ even though the other couplings are of order $1-s$. The two dimensions are

$$\Delta_\pm = 1 + \frac{T}{2} \pm \sqrt{\frac{T^2}{4} - D}, \qquad \Delta_\mathcal{E} = \Delta_-, \tag{4.3}$$

and the square root is $\sqrt{2(1-s)}$ times a power series in $1-s$ [9]. Odd powers of $\tau = \sqrt{x}$ in Eq. (2.2) come from this eigenvalue splitting. In the GFF variable, $1-\epsilon = 4x = 4\tau^2$, so $F(\epsilon)$ has a square-root branch point at $\epsilon = 1$, where $\Delta_\mathcal{E} - 1 \simeq -2\sqrt{\Delta_\sigma}$ and the slope $\mathrm{d}\Delta_\mathcal{E}/\mathrm{d}\Delta_\sigma$ diverges. Continuing around the branch point ($\tau \to -\tau$) exchanges Δ_- and Δ_+. The variable $v = 1 - 2\tau$ removes this branch point: the GFF point $\tau = \frac{1}{2}$ is $v = 0$, the short-range end is $v = 1$, and the image of the GFF point on the second sheet, $\tau = -\frac{1}{2}$, is $v = 2$.

Interpretation. We read Eq. (4.3) as the structure of a fixed-point merger at $s = 1$: the coupling vanishes as $\sqrt{\Delta_\sigma}$, two operators become degenerate at dimension 1, and their mixing is linear in the coupling, so the dimensions split as $\pm\sqrt{\Delta_\sigma}$. This is the pattern of a Kosterlitz–Thouless-type endpoint, consistent with the inverse-square Ising chain at $s = 1$ [20]. In $d = 2, 3$ the short-range end is instead an analytic crossover [10]: a single coupling $g_{\rm SR}$ with $\beta = -\delta g_{\rm SR} + b_0 g_{\rm SR}^3$ has $g^2_{{\rm SR}*} = \delta/b_0$, and since $\gamma_\mathcal{E} = a_0 g^2_{\rm SR} + \ldots$ is even in the coupling, $\Delta_\mathcal{E} = E + (a_0/b_0)\delta + O(\delta^2)$ is analytic in $\delta = x - S$; in $d = 2$ the linear term vanishes and Eq. (2.3) holds. The analytic forms (2.3)–(2.4) are inputs; the merger reading of $d = 1$ is ours.

Consequence for GFF-only resummation. Each finite-order Borel approximant of the ϵ series, even with the Borel-plane singularity built in, is analytic at $\epsilon = 1$ and cannot develop the infinite slope. The GFF-only CB8 curve in $d = 1$ reaches 0.567 at $\Delta_\sigma = 0$ instead of 1. In $d = 2, 3$, where the endpoint is analytic, the same GFF-only prescription reaches 1.0049 at $\Delta_\sigma = \frac{1}{8}$ (exact value 1, core band $[0.944, 1.071]$) and 1.4161 at $\Delta_\sigma = S$ (exact 1.4126, core band $[1.4124, 1.4203]$).

How much analytic structure can the GFF series carry? The test of Table 6 resums the GFF coefficients alone, without short-range input, in increasingly informed variables: ϵ; v, which removes the branch point (with $A_v = A_\epsilon/2$); $\lambda = 2v/(2-v)$, which also sends the mirror point $v = 2$ to infinity; and v with only the exact value $\Delta_\mathcal{E}(0) = 1$ imposed, $F = 1 + (1-v)\mathcal{R}[\cdot]$. Each step moves the center toward the $K = 8$ baseline: at $\Delta_\sigma = 0.1$ the deficit shrinks from -25.6×10^{-3} to -15.8×10^{-3}, -6.0×10^{-3} and $+0.6\times10^{-3}$. Their core bands, however, are 20 to 65 times wider than that of the $K = 8$ baseline at $\Delta_\sigma = 0.1$ (2.0×10^{-3}), and in v and λ they grow rapidly toward $\Delta_\sigma = 0$. Analytic structure helps the center; only the short-range coefficients control the spread.

Δ_σ	$K = 8$ baseline	GFF only, ϵ	GFF only, v	GFF only, λ	v with $\Delta_{\mathcal{E}}(0) = 1$
0.001	0.937424	−371.3 [188.0]	−237.9 [776.2]	−53.3 [3018]	−2.1 [45.7]
0.025	0.713432	−164.5 [144.9]	−104.6 [367.0]	−33.1 [1036]	−0.6 [108.0]
0.05	0.622235	−89.9 [106.0]	−56.4 [212.0]	−19.4 [510.2]	+0.9 [88.2]
0.1	0.530246	−25.6 [47.3]	−15.8 [71.7]	−6.0 [127.6]	+0.6 [42.8]
0.15	0.491712	−5.0 [13.6]	−3.2 [16.8]	−1.5 [21.6]	−0.2 [13.6]
0.2	0.483116	−0.3 [1.2]	−0.2 [1.4]	−0.1 [1.5]	−0.1 [1.3]

Table 6: GFF-only CB8 in different variables. The second column is the $K = 8$ baseline; the other columns give each GFF-only curve minus the baseline, in units of 10^{-3}, with the width of its core band (orders 6–8, $b_{\mathrm{L}} = 2, 3, 4$, $\alpha = -1, 0, 1$) in brackets, in the same units. At $\Delta_\sigma = 0.001$ the four GFF-only values are 0.566, 0.700, 0.884 and 0.935; the exact limit at $\Delta_\sigma = 0$ is 1.

4.5 Scans and what the bands mean

Table 7 lists the prescriptions. The *core band* is the pointwise minimum and maximum over orders $6, 7, 8$, three Borel–Leroy parameters and $\alpha = -1, 0, 1$ (27 approximants). In $d = 1$ the reported *sensitivity envelope* is the union of the core bands for $K = 5, 6, 7, 8$, so it also covers the number of imposed short-range coefficients. It starts at $K = 5$, one coefficient beyond the archived construction. Including $K = 4$ would widen it from at most 4.67×10^{-3} to at most 6.12×10^{-3} (near $\Delta_\sigma = 0.0975$). For $\Delta_\sigma \leq 0.04$ the $K = 4$ curve lies below the $K = 5$–8 envelope (by 5.8×10^{-5} at $\Delta_\sigma = 0.025$, Table 9). Every final HT and converged NRG value lies inside either envelope within its quoted error (Sec. 6.2). Wider scans (b_{L} from 0 to 6 or 7, α from -2 to 2) and action variations ($1/A$ scaled by 0.8, 0.9, 1.1, 1.2) are archived as diagnostics. No HT, NRG or MC value was used to choose these settings. In $d = 1$, however, the number $K = 8$ of imposed short-range coefficients, and with it the range $K = 5$–8 of the envelope, was fixed after these data had been seen (Sec. 5.2).

d	Baseline	Core b_{L}	PB control shown	Action in ϵ
1	CB8, endpoint-v, $K = 8$, $b_{\mathrm{L}} = 3$, $\alpha = 0$	$2, 3, 4$	PB[5/3], $b_{\mathrm{L}} = 3$ ($K \leq 7$)	0.5140649
2	CB8, curvature, $b_{\mathrm{L}} = 3.5$, $\alpha = 0$	$2.5, 3.5, 4.5$	PB8 [5/3], $b_{\mathrm{L}} = 3.5$	9/8
3	CB8, slope, $b_{\mathrm{L}} = 4$, $\alpha = 0$	$3, 4, 5$	PB7 [3/4], $b_{\mathrm{L}} = 4$	1.9234333

Table 7: Baseline prescriptions and core scans. The core scan always uses orders $6, 7, 8$ and $\alpha = -1, 0, 1$. In $d = 1$ PB[5/3] is admissible only for $K \leq 7$; in $d = 3$ every eighth-order PB candidate at $b_{\mathrm{L}} = 4$ is rejected, so the seventh-order control is shown.

5 Results

5.1 From three loops to eight loops and the endpoint

Table 8 shows, in each dimension, how the curve changes as information is added: CB3 uses only $C_0, \ldots, C_3$, the coefficients known before this work [7], in the GFF-only prescription; CB8 adds $C_4, \ldots, C_8$; the baseline adds the short-range end. In $d = 1$ the GFF series alone is far from the answer over most of the window. In $d = 2$ the eighth-order GFF-only curve is already within 5.2×10^{-3} of the baseline everywhere (5.24×10^{-3} at $\Delta_\sigma = 0.193$), and in $d = 3$ within 3.4×10^{-3} (at the endpoint; 1.6×10^{-3} at $\Delta_\sigma = 0.55$), while CB3 is off by up to 0.087 in $d = 2$.

d	Δ_σ	CB3, GFF only	CB8, GFF only	CB8 + short-range end (baseline)
1	0.05	0.458989	0.532338	0.622235
1	0.1	0.463610	0.504636	0.530246
1	0.15	0.470013	0.486725	0.491712
1	0.2	0.479926	0.482857	0.483116
2	0.125	0.913472	1.004921	1 (imposed)
2	0.2	0.921916	0.980233	0.985419
2	0.3	0.936589	0.960577	0.961182
2	0.4	0.958418	0.962702	0.962567
3	S	1.410634	1.416062	1.412625 (imposed)
3	0.55	1.418920	1.422866	1.421232
3	0.6	1.433555	1.435599	1.435241
3	0.65	1.450835	1.451576	1.451541

Table 8: Progression of the resummed $\Delta_{\mathcal{E}}$ with the information used. GFF-only curves resum F directly in ϵ with $\alpha = 0$ and the baseline b_{L} of each dimension.

5.2 One dimension

Figure 1 shows the baseline over the full interval and its residuals with respect to the comparison data, and Table 9 gives representative values. The curve falls from the imposed limit 1 at $\Delta_\sigma = 0$, reaches a shallow minimum 0.482813 at $\Delta_\sigma = 0.192$ and rises to the GFF value $\frac{1}{2}$. The sensitivity envelope is 1.45×10^{-3}, 3.64×10^{-3} and 4.40×10^{-3} wide at $\Delta_\sigma = 0.05$, 0.1, 0.15 and at most 4.67×10^{-3} wide (at $\Delta_\sigma = 0.134$); it is not drawn in the figures. The coefficient-allowance budget is at most 6.0×10^{-5} (at $\Delta_\sigma = 0.141$), far below it.

Δ_σ	Baseline CB8	$K = 5$–8 envelope	$K = 4$ (archived)	endpoint-ϵ, $K = 8$	GFF only
0.025	0.713432	[0.713301, 0.713615]	0.713243	0.713437	0.548904
0.05	0.622235	[0.621524, 0.622977]	0.621623	0.622296	0.532338
0.075	0.566627	[0.565044, 0.567868]	0.565583	0.566838	0.517472
0.1	0.530246	[0.527952, 0.531597]	0.528948	0.530651	0.504636
0.125	0.506428	[0.503992, 0.508580]	0.505179	0.506948	0.494227
0.15	0.491712	[0.489809, 0.494205]	0.490810	0.492177	0.486725
0.175	0.484240	[0.483202, 0.486233]	0.483803	0.484506	0.482703
0.2	0.483116	[0.482766, 0.484000]	0.483012	0.483189	0.482857
0.225	0.488085	[0.488058, 0.488177]	0.488081	0.488088	0.488077

Table 9: One-dimensional curves: CB8 with $b_{\mathrm{L}} = 3$, $\alpha = 0$ and the final $C_8(1)$. The envelope column is the union of the core bands for $K = 5$–8; the endpoint columns impose $C_1^{\mathrm{SR}}, \ldots, C_K^{\mathrm{SR}}$; the last column is the GFF-only curve.

Dependence on the number of short-range coefficients. The short-range input is the most important ingredient after the GFF series itself. The core band narrows from $K = 4$ to $K = 7$ and widens slightly at $K = 8$: at $\Delta_\sigma = 0.1$ its width is 6.11×10^{-3}, 3.64×10^{-3}, 2.33×10^{-3}, 1.52×10^{-3} and 2.00×10^{-3} for $K = 4, \ldots, 8$, and at $\Delta_\sigma = 0.05$ it falls from 3.4×10^{-3} ($K = 4$) to 2.3×10^{-4} ($K = 8$). The central value moves with K, non-monotonically, by up to 1.08×10^{-3} among $K = 5, \ldots, 8$ (near $\Delta_\sigma = 0.121$) and by up to 1.32×10^{-3} from $K = 4$ to $K = 8$. This K dependence is part of the envelope. Replacing C_8^{SR} by the independent value 242.83 moves the curve by at most 3×10^{-5}.

Endpoint variable and Padé controls. At $K = 8$ the endpoint-ϵ control differs from the v baseline by at most 5.2×10^{-4} (at $\Delta_\sigma = 0.13$) and stays inside the envelope. The matched PB[5/3]

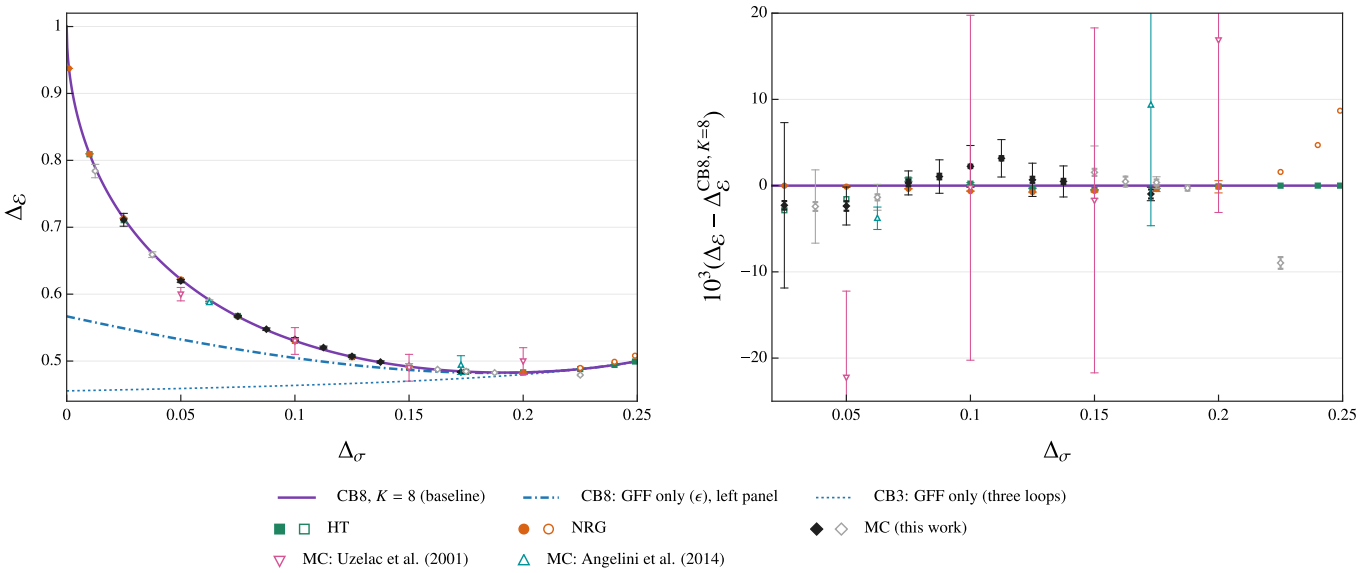


Figure 1: One dimension. Left: $\Delta_\mathcal{E}$ over the full interval. Purple: baseline (CB8, endpoint-v, $K = 8$). Blue dash-dot: GFF-only CB8. Blue dotted: CB3, the GFF-only curve from $C_0, \ldots, C_3$. Right: residuals $10^3(\Delta_\mathcal{E} - \Delta_\mathcal{E}^{\mathrm{baseline}})$ for $0.02 \le \Delta_\sigma \le 0.25$. Green squares: HT [4] (filled: final; at $\Delta_\sigma = 0.075$ one-sided, with a dashed extra bar of -5×10^{-4}; open: unstable or no limit, value at cutoff $N = 40$). Orange circles: NRG [5] (open: not converged in chain depth, no error, not compared). Diamonds, labeled "MC (this work)" as in Ref. [1]: the project's MC [6] (filled black: accepted; open light gray: other labels), with separate statistical and systematic bars. Triangles: published MC with quoted errors, one color per reference (rose, inverted: Uzelac et al. [21]; teal: Angelini et al. [22]); the right panel extends to -25 and $+20$ for them, and three of their bars are cut at the frame (full bars in Fig. 3). Labels and errors as in Table 12. The envelope and the PB controls are not drawn.

control ($b_\mathrm{L} = 3$, same F_K^{SR} and variable) is admissible only for $K = 4, \ldots, 7$; at $K = 8$ and order 8 only [3/5] survives the pole screen, and no $K = 8$ PB curve is substituted. The matched CB−PB difference changes sign and size with K; its signed extreme is -5.5×10^{-4}, $+3.4 \times 10^{-4}$, -4.6×10^{-5} and -1.5×10^{-4} for $K = 4, 5, 6, 7$ (Fig. 4, top). The CB/PB comparison is therefore a comparison of families, not evidence for either.

The choice $K = 8$. We impose $K = 8$ because these are all the short-range coefficients we have. This choice, and the range $K = 5$–8 of the envelope, were not fixed before the comparison data had been seen. Choosing the K with the narrowest core band (with an admissible PB control) would give $K = 7$, and the comparison data do not favor $K = 8$ over $K = 4$ (Sec. 6.2). Whether the envelope should also include the choice of endpoint variable is open; at present the endpoint-ϵ curve lies inside it.

5.3 Two and three dimensions

Figure 2 shows the baselines with the GFF-only curves and the comparison data; Tables 10 and 11 give values.

Two dimensions. The curvature baseline has a minimum $\Delta_\mathcal{E} = 0.958072$ at $\Delta_\sigma = 0.346$. Adding C_8 changed the seventh-order curve by at most 8.29×10^{-4}, and the matched PB8 control ([5/3], $b_\mathrm{L} = 3.5$) differs from CB8 by at most 7.88×10^{-4}; at $b_\mathrm{L} = 0$ every eighth-order PB candidate has a positive-real pole. The propagated coefficient allowance is at most 7.75×10^{-5}. The largest uncertainty is the endpoint construction (Table 10). The tangent and value constructions differ from the baseline by up to 5.7×10^{-3} and 7.5×10^{-3}. The tangent construction lies outside the baseline's core band for all $\Delta_\sigma < 0.476$, the value construction for $\Delta_\sigma < 0.26$; for $\Delta_\sigma < 0.29$ the

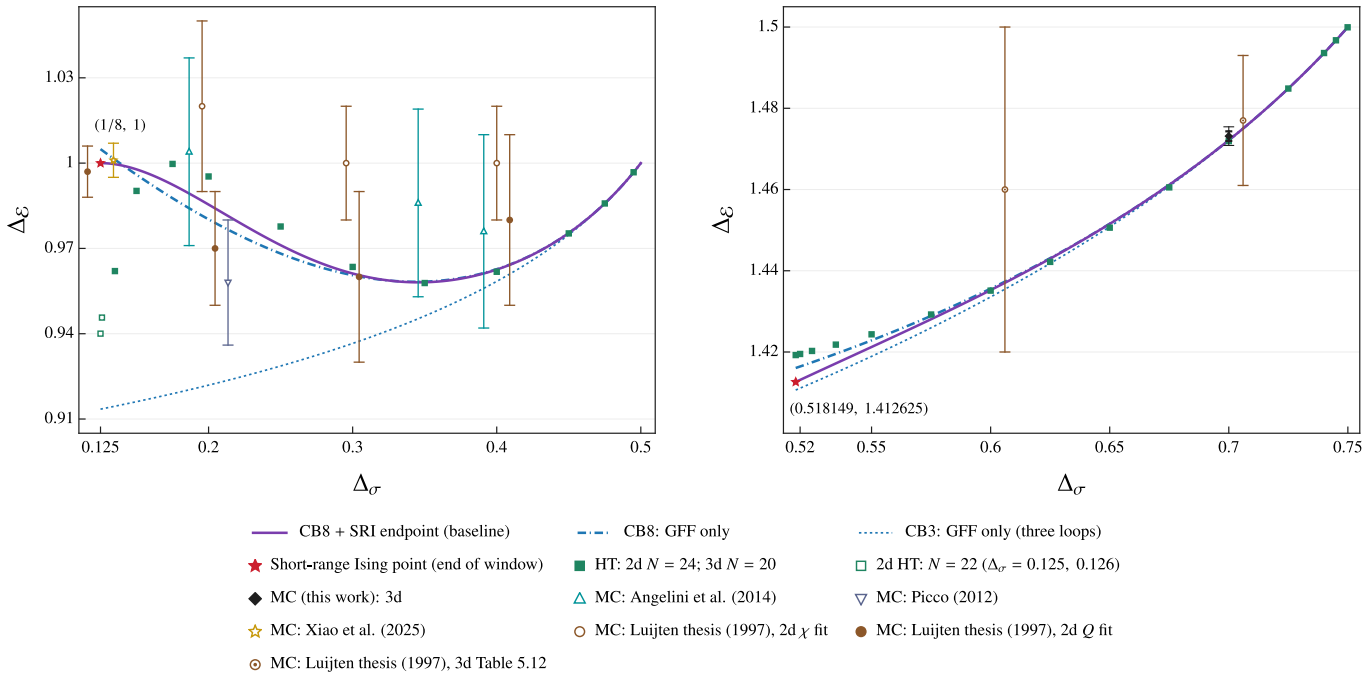


Figure 2: Two dimensions (left) and three dimensions (right). Purple: baseline. Blue dash-dot: GFF-only CB8. Blue dotted: CB3. Green squares: finite-cutoff HT [4] (2d $N = 24$; open: $N = 22$ at $\Delta_\sigma = 0.125, 0.126$, where the tuning has no solution at $N = 24$; 3d $N = 20$), without error bars. Black diamond: the project's MC value in 3d [6], accepted only under amendment P1 (Table 12). Red star: the short-range Ising point, where the baseline is pinned: (1/8, 1) in 2d and (0.518149, 1.412625) in 3d [11]. Other markers: published MC with quoted errors, one color and shape per reference (2d: Angelini et al. [22], teal triangles; Picco [24], slate inverted triangle; Xiao et al. [25], ochre star; the χ (open) and Q (filled) estimators of Luijten's thesis [23], brown circles; 3d: Luijten's thesis, brown circles with a central dot), shifted horizontally by at most 0.0135 (2d) and 0.006 (3d) where they share a Δ_σ. Not drawn: the core bands of the baselines (within -4.07×10^{-3} and $+1.23 \times 10^{-3}$ of the 2d curve, -7.24×10^{-5} and $+8.09 \times 10^{-5}$ of the 3d curve); the 2d value and tangent constructions; the value of Angelini et al. at $s = \frac{7}{4}$ (1.02 ± 0.10).

Δ_σ	curvature CB8	core band	tangent CB8	value CB8	GFF-only CB8	PB8 [5/3]
0.15	0.997797	[0.997747, 0.997823]	0.998560	0.992255	0.995913	0.997816
0.2	0.985419	[0.984485, 0.985834]	0.989418	0.978404	0.980233	0.985708
0.25	0.971143	[0.968443, 0.972158]	0.976830	0.967441	0.968198	0.971821
0.3	0.961182	[0.957192, 0.962396]	0.965863	0.960349	0.960577	0.961953
0.35	0.958099	[0.954550, 0.958909]	0.960426	0.958296	0.958332	0.958584
0.4	0.962567	[0.960930, 0.962810]	0.963108	0.962704	0.962702	0.962702
0.45	0.975429	[0.975260, 0.975440]	0.975448	0.975436	0.975436	0.975434

Table 10: Two dimensions: the three endpoint constructions and the GFF-only curve (CB8, $b_{\mathrm{L}} = 3.5$, $\alpha = 0$), the core band of the baseline, and the PB control PB8 [5/3] of the curvature construction.

larger of the two deviations exceeds the width of the core band. Varying $1/A_\epsilon$ by $\pm 20\%$ moves the baseline by at most 1.7×10^{-3}.

Δ_σ	slope CB8	core band	value CB8	GFF-only CB8	PB7 [3/4]	PB8 [5/3]
0.55	1.421232	[1.421196, 1.421266]	1.421119	1.422866	1.421226	1.421225
0.575	1.428068	[1.428002, 1.428137]	1.427959	1.428886	1.428059	1.428057
0.6	1.435241	[1.435170, 1.435321]	1.435165	1.435599	1.435231	1.435229
0.65	1.451541	[1.451512, 1.451582]	1.451526	1.451576	1.451537	1.451537
0.7	1.472143	[1.472141, 1.472146]	1.472142	1.472143	1.472143	1.472143

Table 11: Three dimensions: the slope (baseline) and value constructions and the GFF-only curve (CB8, $b_{\mathrm{L}} = 4$, $\alpha = 0$), the core band of the baseline, and PB controls of the slope construction: PB7 at $b_{\mathrm{L}} = 4$ and PB8 at $b_{\mathrm{L}} = 0$, since at order 8 and $b_{\mathrm{L}} = 4$ every PB candidate is rejected.

Three dimensions. The slope baseline increases monotonically across the window. Adding C_8 changed the seventh-order curve by at most 1.13×10^{-5}; the admissible PB8 at $b_{\mathrm{L}} = 0$ ([5/3]) differs by at most 1.21×10^{-5}, and the propagated coefficient allowance is at most 1.66×10^{-5}. The estimated range of the short-range slope, $c_1 \in [0.269, 0.276]$ (status conditional), moves the curve by at most 6.97×10^{-5}, comparable to the width of the core band; the value construction stays within 1.2×10^{-4} of the baseline. In $d = 3$ the resummation is therefore tightly constrained; its uncertainty is dominated by the short-range inputs and by assumptions shared by all prescriptions.

6 Comparison with HT, NRG and Monte Carlo

6.1 The comparison data and their status

The comparison values come from the HT [4], NRG [5] and MC [6] notes of the release of Ref. [1] and from the literature; frozen copies ship with this note (Appendix B). None of them enters a curve. Their uncertainties have different meanings, summarized in Table 12, and their labels (final, converged, accepted and so on) were assigned in those notes by their own criteria, not by comparison with the curves of this note. The errors must not be combined as independent Gaussian standard deviations: below, a difference is compared with each quoted error separately or with the linear sum of the errors involved, never with a quadrature sum, and the ratios are descriptive, not χ^2 statistics.

6.2 One dimension

Figure 3 shows all 48 displayed comparison points relative to the baseline (the unconverged HT point at $\Delta_\sigma = 0.001$ is omitted). With $K = 8$:

- The final HT values lie within -5.0×10^{-4} to $+2.3 \times 10^{-4}$ of the baseline for $0.1 \le \Delta_\sigma \le 0.249$; the one-sided point at $\Delta_\sigma = 0.075$ is $+7.0 \times 10^{-4}$ above.
- The final NRG values ($0.001 \le \Delta_\sigma \le 0.2$) lie 1×10^{-7} to 7.6×10^{-4} below the baseline, the largest offset at $\Delta_\sigma = 0.125$.
- The nine accepted MC values lie within -2.4×10^{-3} to $+3.2 \times 10^{-3}$.

The $K = 5$–8 envelope contains every final and one-sided HT value and every final NRG value within its quoted error; all but two lie inside it, and HT at $\Delta_\sigma = 0.249$ and NRG at 0.001 lie 4×10^{-8} and 9×10^{-8} outside. Every accepted MC value reaches the envelope within the sum of its statistical error and σ_{sys}; the largest distance is 0.60 of that sum (at $\Delta_\sigma = 0.05$). All of this also holds for the envelope extended to $K = 4$. Among the points with other labels, the conditional MC

Data	Meaning and status
HT, $d = 1$	Hamiltonian truncation corrected for the truncation artifact, with the zero mode in a density-matrix basis, extrapolated to infinite cutoff from $N \leq 40$ [4]. *Final* ($0.1 \leq \Delta_\sigma \leq 0.249$): a stable extrapolation; error = half the spread of the accepted fits, a model spread. *One-sided* ($\Delta_\sigma = 0.075$): the local exponent still rises, so the fits may overestimate the limit; extra allowance -5×10^{-4}. *Unstable* ($\Delta_\sigma = 0.05$): the fit families disagree; no value quoted. *No limit* ($\Delta_\sigma \leq 0.025$): the corrected sequence turns over and keeps falling; the value at $N = 40$ is shown. The error does not include the dependence on how the artifact is corrected (Sec. 6.2).
HT, $d = 2, 3$	Finite-cutoff values only [4] (2d: $N = 24$, $N = 22$ near $\Delta_\sigma = \frac{1}{8}$; 3d: $N = 20$). Solver residuals and last cutoff steps are not continuum error bars.
NRG, $d = 1$	Bosonic NRG of the sub-ohmic spin–boson model [5]: the settled value of the flow, a plateau or (at $\Delta_\sigma = 0.175$, 0.2) an extrapolation in chain depth. Error: chain depth, spread over the numbers of kept states and oscillator levels, and discretization, added linearly; not a standard deviation. *Final* (converged; $0.001 \leq \Delta_\sigma \leq 0.2$); *not converged in chain depth* ($\Delta_\sigma = 0.225$, 0.24, 0.249, near $s = \frac{1}{2}$): settled value shown without error, not compared.
Project MC	Cluster MC of the lattice model [6]; σ_{stat} is a bootstrap error, σ_{sys} half the range of the values over a family of admissible fits or extrapolations, a sensitivity scale, not a standard deviation. Labels from a fixed decision table: *accepted*, the value passed every check of the fit protocol (σ_{sys}*-dominated*: with σ_{sys} more than three times the median σ_{stat}); *conditional*, a single extrapolation, so no σ_{sys}; *cross-check unavailable*, fewer than two independent extrapolations pass their gates; *cross-check failed*, two independent extrapolations differ by more than twice the error of their difference; *model-limited*, no fit passes. $d = 1$: 17 values, 9 accepted. $d = 3$: one value, at $\Delta_\sigma = 0.7$, accepted only under amendment P1, which re-profiles the correction exponent in every bootstrap draw of σ_{stat} and was adopted after the protocol had been applied to the production data. Where no infinite-size value exists, the finite-size exponent $\Delta_{\text{eff}}(U)$ of the two largest lattices, from the slopes of the Binder cumulant U at its crossing for sizes L and $2L$, is shown.
Published MC	Quoted errors. $d = 1$: Uzelac et al. [21] (4 points), Angelini et al. [22] (2). $d = 2$: 13 values from Refs. [22, 23, 24, 25]. $d = 3$: Luijten's thesis [23] (2).
Bootstrap, FRG	$d = 3$ only, read off published figures [26, 27]; we assign ± 0.005 to the bootstrap kinks; not plotted.

Table 12: Comparison data, their uncertainty conventions and labels.

value at $\Delta_\sigma = 0.225$ lies 9.0×10^{-3} below, and the NRG points not converged in chain depth rise to 8.7×10^{-3} above at $\Delta_\sigma = 0.249$. Of the published MC values, four lie within one quoted error of the envelope, while Uzelac et al. at $\Delta_\sigma = 0.05$ and Angelini et al. at $\Delta_\sigma = 0.0625$ lie 2.2 and 2.0 quoted errors below it.

Agreement among the methods. Away from the GFF end, HT, NRG and MC agree at the sub-percent level. The largest relative difference among accepted MC, final HT and converged NRG values at a common coupling is 0.54%, between MC and NRG at $\Delta_\sigma = 0.1$, where HT and MC differ by 0.38%. HT and NRG agree to within two parts in a thousand wherever HT has a continuum limit ($0.075 \le \Delta_\sigma \le 0.2$; the largest relative difference, 1.9×10^{-3}, is at the one-sided point $\Delta_\sigma = 0.075$). Near the GFF end the MC values at finite size lie 0.006 to 0.010 below the resummed curve: the exponents $\Delta_{\mathrm{eff}}(U)$ of the largest lattice pair at $\Delta_\sigma = 0.2125$ and 0.2375 (Table 18) and the conditional fit at 0.225. There the ϵ series is accurate and these MC values are not infinite-size estimates.

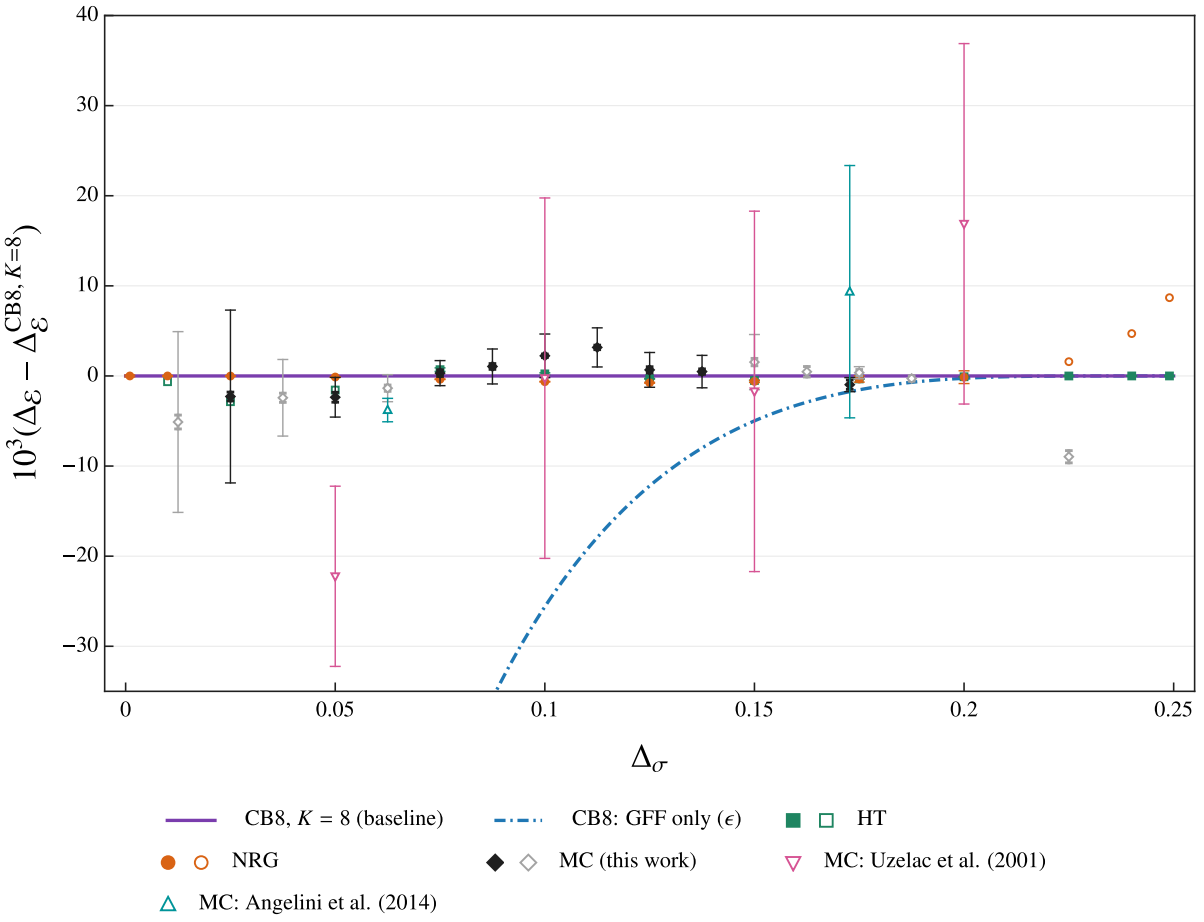


Figure 3: One dimension: deviations $10^3(\Delta_{\mathcal{E}} - \Delta_{\mathcal{E}}^{\mathrm{baseline}})$ of all displayed reference points from the $K = 8$ baseline, with full error bars, including the published MC values (rose inverted triangles: Uzelac et al. 2001; teal triangles: Angelini et al. 2014). Markers as in Fig. 1. Blue dash-dot: GFF-only CB8, below the frame for $\Delta_\sigma < 0.088$ (it reaches -433 at $\Delta_\sigma = 0$). The $K = 5$–8 envelope is not drawn; its width is at most 4.67×10^{-3}, that is 4.67 in these units.

HT and NRG. At $\Delta_\sigma = 0.15$–0.2 the final HT and NRG values agree (differences at most 1.0×10^{-4}, below the sum of their errors). At $\Delta_\sigma = 0.1$ and 0.125 the NRG values lie 8.7×10^{-4} and 4.5×10^{-4} below the HT values, 5.8 and 4.6 times the sum of the HT fit half-spread and the NRG error. The gap is of the size of the dependence of HT on the correction of its truncation artifact: without the correction the truncation extrapolates 2.3×10^{-3} and 6.6×10^{-4} lower at these

points (1.7×10^{-3} at 0.075), and NRG lies between the two limits. The HT note does not combine this dependence into its error, because the truncation has no strict error bar and an error built after the gap to NRG had been seen would make the two methods appear to agree [4]. The HT values therefore keep their fit half-spreads, and the difference between the corrected and uncorrected limits is shown separately.

Do the data select $K = 8$ over $K = 4$? The archived $K = 4$ and the baseline $K = 8$ curves differ by at most 1.32×10^{-3} (at $\Delta_\sigma = 0.109$). That is less than the width of the $K = 8$ core band at the same points (2.0×10^{-3} at $\Delta_\sigma = 0.1$, 3.0×10^{-3} at 0.125), so a preference between them is a choice inside the prescription dependence. Table 13 compares the separation with the quoted errors and lists the dependence of HT on the artifact correction separately.

- *MC* does not resolve the two curves anywhere: the separation is at most 0.59 of the sum of σ_{stat} and σ_{sys} of an accepted value.
- *HT.* At $\Delta_\sigma = 0.075$ and 0.1 HT lies above both curves, nearer $K = 8$, but its dependence on the artifact correction (1.7×10^{-3}, 2.3×10^{-3}) exceeds the separation of the curves (1.0×10^{-3}, 1.3×10^{-3}), so HT does not decide there. At 0.125 it lies between the curves, nearer $K = 8$. At 0.15–0.2, where the prescription dependence is small ($\leq 1.3 \times 10^{-4}$), it lies between the curves and nearer $K = 4$: 1×10^{-5} to 4×10^{-4} above $K = 4$ (2.6–12 errors) and 9×10^{-5} to 5×10^{-4} below $K = 8$ (15–21 errors).
- *NRG* selects $K = 8$ only at $\Delta_\sigma = 0.025$, where the separation is 1.9×10^{-4} and $K = 4$ is 6.2 errors away. At 0.05–0.15 it lies between the curves, more than 2.5 errors from each: nearer $K = 8$ at 0.05, 0.075 and, marginally, 0.1 (19.1 against 19.7 errors), nearer $K = 4$ at 0.125 and 0.15. At 0.175, 0.2 and $\Delta_\sigma \leq 0.01$ the separation is below two errors.

So the precise data resolve the $K = 4$–$K = 8$ difference at several points, but no method favors $K = 8$ over its range. Where both curves are resolved, HT and NRG mostly lie between them, and at 0.15–0.2, where HT is most precise and depends least on the artifact correction, they lie nearer $K = 4$. An earlier version of this note justified $K = 8$ by its agreement with HT and NRG. That argument relied on the HT fit half-spread alone at $\Delta_\sigma = 0.1$–0.125 and on earlier NRG values; it does not survive the dependence of HT on the artifact correction, which exceeds the separation at $\Delta_\sigma = 0.075$ and 0.1, and it is withdrawn. The choice $K = 8$ rests only on its a priori motivation: these are all the short-range coefficients we have, and at fixed K the prescriptions then spread much less than with $K = 4$ (core band at $\Delta_\sigma = 0.05$: 2.3×10^{-4} against 3.4×10^{-3}) and depend less on b_{L} (Secs. 5.2 and 7). It was fixed after the comparison data had been seen, and taken together these data do not favor it over $K = 4$.

6.3 Two dimensions

The finite-cutoff HT values [4] at the largest cutoff ($N = 24$ at eleven couplings, $N = 22$ at $\Delta_\sigma = 0.125$ and 0.126) compare with the curvature baseline as follows (Fig. 2, left):

- At weak coupling, $\Delta_\sigma \geq 0.35$ ($s \leq 1.3$), HT agrees with CB8 to at most 8×10^{-4}, inside the core band.
- In mid-window HT lies above the baseline: $+2.3 \times 10^{-3}$ at $\Delta_\sigma = 0.3$, $+6.6 \times 10^{-3}$ at 0.25, $+9.9 \times 10^{-3}$ at 0.2 and $+7.3 \times 10^{-3}$ at 0.175. At $s = 1.4$–1.5 ($\Delta_\sigma = 0.3$–0.25) HT lies within the spread of the endpoint constructions, inside the core band of the tangent construction. At $s = 1.6$–1.65 ($\Delta_\sigma = 0.2$–0.175) HT is above every construction, by about 3×10^{-3} above the upper edge of the tangent core band. The cutoff dependence does not show whether this excess survives at infinite cutoff. At $s = 1.4$ and 1.5 HT rises monotonically with N, so if the rise continues the $N = 24$ values are lower bounds. At $s = 1.6$, however, it is flat from $N = 18$ on and falls by 5×10^{-5} in the last step, and at $s = 1.65$ it falls steadily from $N = 10$

		HT (corrected)					NRG			
Δ_σ	$10^4(F_4-F_8)$	$10^4(y-F_8)$	err	z_8, z_4	corr.−unc.		$10^4(y-F_8)$	err	z_8, z_4	
0.025	−1.89	·	·	·	·		−0.10	2.9×10^{-5}	−0.4, +6.2	near 8
0.05	−6.12	·	·	·	·		−1.19	2.7×10^{-5}	−4.5, +18.4	b(8)
0.075	−10.43	+6.99	2.0×10^{-4}	+3.5, +8.8	1.7×10^{-3}	o(8)†	−3.67	3.0×10^{-5}	−12.3, +22.7	b(8)
0.1	−12.99	+2.31	1.2×10^{-4}	+2.0, +13.2	2.3×10^{-3}	o(8)†	−6.39	3.3×10^{-5}	−19.1, +19.7	b(8)
0.125	−12.48	−3.09	4.3×10^{-5}	−7.2, +21.8	6.6×10^{-4}	b(8)	−7.55	5.4×10^{-5}	−14.1, +9.2	b(4)
0.15	−9.01	−5.03	3.3×10^{-5}	−15.4, +12.2	1.3×10^{-4}	b(4)	−6.06	1.1×10^{-4}	−5.3, +2.6	b(4)
0.175	−4.37	−3.32	1.5×10^{-5}	−21.4, +6.8	3.3×10^{-5}	b(4)	−3.40	3.5×10^{-4}	−1.0, +0.3	n.r.
0.2	−1.04	−0.92	4.7×10^{-6}	−19.4, +2.6	4.5×10^{-6}	b(4)	−1.36	7.1×10^{-4}	−0.2, −0.0	n.r.

Table 13: Can the comparison data tell the $K = 4$ and $K = 8$ curves apart? F_K: CB8 with the short-range series through C_K^{SR}; y: the HT or NRG value; $z_K = (y - F_K)/\mathrm{err}$. HT err: half-spread of the accepted fits (quoted); at $\Delta_\sigma = 0.075$ its one-sided allowance -5×10^{-4} is not included and would lower z_8 to 1.0. corr.−unc.: the HT limit corrected for the truncation artifact minus the uncorrected limit, shown separately and not combined with err (Sec. 6.2). NRG err: chain-depth, kept-state and discretization errors added linearly. Labels: near 8, within one error of F_8 and more than two from F_4; b(K), between the curves, and o(K), outside both, more than one error from each and nearer F_K; n.r., separation below two errors. †: the dependence on the artifact correction exceeds the separation. MC resolves neither curve anywhere. The errors are not standard deviations; the ratios are descriptive.

on (by 5.3×10^{-3} up to $N = 24$, 1.0×10^{-3} in the last step), so larger cutoffs could reduce the excess there.

- Near the short-range end ($\Delta_\sigma \leq 0.15$) HT drifts downward, away from the exact endpoint value 1 (0.940 at $\Delta_\sigma = \frac{1}{8}$, $N = 22$), and is not usable.

The discrepancy is not assigned to either method: the cutoff dependence and the artifact correction of HT, the uncomputed orders and the endpoint assumptions all remain. The thirteen published MC values have quoted errors from 0.006 to 0.034 (and 0.10 for Angelini et al. at $s = \frac{7}{4}$). Away from $s = \frac{7}{4}$ they lie within two quoted errors of both CB8 and HT, so they cannot separate them; at $s = \frac{7}{4}$ they agree with the exact value 1, which HT misses. Luijten's two estimators on the same data differ by up to 0.05. The project's own 2d MC values are conditional or model-limited [6], so only finite-size exponents are available (Table 14). The exponents $\Delta_{\mathrm{eff}}(U_4)$ of the largest lattice pair ($L = 512 \to 1024$) agree with the curvature baseline to 1.5×10^{-3} at $s = 1.2$–1.4. There the endpoint constructions differ from it by up to 4.7×10^{-3}, so the exponents do not select one. Toward both ends ($s = 1.05$, 1.1 and 1.5–1.7) they fall short of the baseline by 0.005 to 0.02.

s	Δ_σ	label	$\Delta_{\mathrm{eff}}(U_4)$	curvature CB8	10^3(MC−CB8)	10^3 max |constr. − CB8|
1.05	0.475	conditional	0.9702(25)	0.985836	−15.65	0.00
1.1	0.45	conditional	0.9659(23)	0.975429	−9.51	0.02
1.2	0.4	conditional	0.9631(22)	0.962567	+0.56	0.54
1.3	0.35	model-limited	0.9584(20)	0.958099	+0.28	2.33
1.4	0.3	model-limited	0.9627(20)	0.961182	+1.54	4.68
1.5	0.25	model-limited	0.9661(18)	0.971143	−5.03	5.69
1.6	0.2	model-limited	0.9698(18)	0.985419	−15.66	7.02
1.7	0.15	model-limited	0.9773(17)	0.997797	−20.49	5.54

Table 14: Two dimensions: finite-size exponents $\Delta_{\mathrm{eff}}(U_4)$ of the project's MC for the largest lattice pair, $L = 512 \to 1024$, with their statistical errors [6], against the curvature baseline. Last column: the largest deviation of the tangent and value constructions and the GFF-only curve from the baseline. These are not infinite-size values.

6.4 Three dimensions

Against the slope baseline (Fig. 2, right):

- HT at $N = 20$ is within 1×10^{-3} of CB8 for $\Delta_\sigma \geq 0.6$ ($s \leq 1.8$), for example -2.2×10^{-4} at $\Delta_\sigma = 0.7$ and -9.3×10^{-4} at $\Delta_\sigma = 0.65$. Toward the short-range end it lies above, by 1.2×10^{-3} at $\Delta_\sigma = 0.575$ rising to 6.6×10^{-3} at $\Delta_\sigma = S$, where HT gives 1.41924 against the exact $E = 1.41262528$; as in $d = 2$, HT fails near the short-range end, here lying above the curve rather than below it.
- The project's MC value at $s = 1.6$ [6], $\Delta_\mathcal{E} = 1.47314 \pm 0.00118\,(\text{stat}) \pm 0.00230\,(\text{sys})$, lies 1.0×10^{-3} above CB8 (1.47214). It is accepted only under amendment P1 (Table 12), adopted after the protocol had been applied to the production data; σ_{sys} is the half-range of the values over the admissible extrapolation series.
- The project's extrapolations that are not accepted, at $s = 1.7$, 1.85 and 1.9 (cross-check failed or unavailable), lie 0.022 to 0.033 above CB8.
- Luijten's thesis gives 1.477(16) at $s = 1.6$ and 1.46(4) at $s = 1.8$, +0.3 and +0.6 quoted errors from CB8; Luijten calls these results rather poor (lattices up to $L = 64$, correction exponent fixed by hand).
- Bootstrap kinks read off Fig. 11 of Ref. [26] (conjectured identification) give 1.470, 1.450, 1.432, 1.415 at $s = 1.593$, 1.685, 1.778, 1.871 (read-off error about ± 0.005), below CB8 by 0.004, 0.004, 0.007, 0.010. The gap grows toward the short-range end, where CB8 imposes $c_1 = 0.273$; the kink at $s = 1.871$ implies a smaller slope. A functional-RG truncation [27] departs by up to 0.03 near the short-range end, where its thermal exponent, about 1.543 as read off its Fig. 2, stays below the 3d Ising value 1.5874.

7 Robustness

7.1 Held-out coefficients

A finite-order test that uses no comparison data is to withhold the last known coefficient, construct the approximant from the lower ones and re-expand it to predict the withheld one. This tests finite-order predictive power, not asymptotic dominance. Tables 15 and 16 collect the results; Fig. 4 shows the one-dimensional ones together with the matched CB−PB differences.

Construction ($b_\text{L} = 3$)	C_6	C_7	C_8
GFF only, CB	11.03	1.58	1.55
GFF only, PB ([3/2], [4/2], [5/2])	5.70	0.44	0.33
Endpoint-v, $K = 8$, CB	10.95	5.93	0.94
Endpoint-v, $K = 8$, PB ([4/1], [4/2], [3/4])	3.86	0.17	0.50
Endpoint-v, $K = 4$, CB	7.23	0.37	0.81
Endpoint-v, $K = 4$, PB ([3/2], [4/2], [5/2])	2.53	0.16	0.44

Table 15: One dimension: absolute relative errors (%) of held-out coefficients. The targets C_6, C_7 are unchanged; for C_8 the final value is the target, and the predictions themselves do not depend on it.

Different orders, dimensions and constructions favor different families. For the newly measured C_8, the GFF-only PB prediction is closer in $d = 1$ and $d = 2$, CB is closer in $d = 3$. In $d = 1$ the $K = 8$ CB prediction of C_8 is 0.94% off, which is 4.4 times the allowance of $C_8(1)$ (0.21%). An accurate PB prediction of C_8 also does not guarantee an admissible eighth-order PB curve of the matched shape once C_8 is supplied: at $K = 8$ in $d = 1$ only [3/5] survives at eighth order, and in $d = 3$ no eighth-order candidate survives at $b_\text{L} = 4$. These tests support keeping both families and

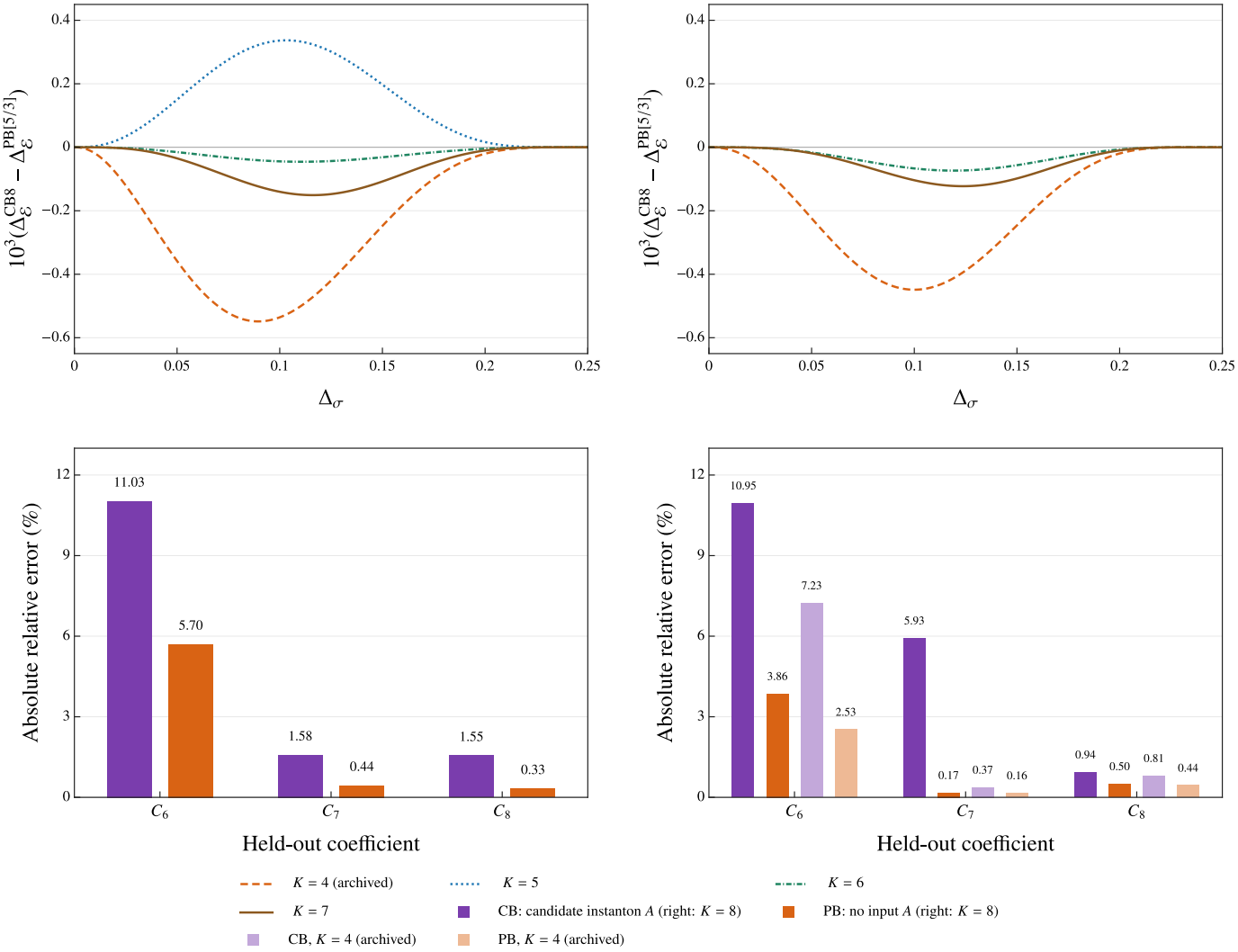


Figure 4: One dimension: conformal Borel against Padé–Borel. Top: matched differences $10^3(\Delta_{\mathcal{E}}^{\mathrm{CB8}} - \Delta_{\mathcal{E}}^{\mathrm{PB}[5/3]})$ with the same coefficients, endpoint input F_K^{SR}, order, variable and $b_{\mathrm{L}} = 3$; left in v for $K = 4$ (archived), 5, 6, 7; right in ϵ for $K = 4, 6, 7$ (where PB[5/3] is admissible). Bottom: absolute relative errors of held-out C_6, C_7, C_8 predicted from the preceding coefficients; left for the GFF-only series (independent of K), right for the endpoint-v construction with $K = 8$ (dark) and $K = 4$ (light). PB shapes as selected by the rule of Sec. 4.1 ($K = 8$: [4/1], [4/2], [3/4]; $K = 4$ and GFF-only: [3/2], [4/2], [5/2]). The C_8 bars use the final $C_8(1)$ (allowance about 0.21%).

	$d = 2$				$d = 3$			
	GFF only		with SR input		GFF only		with SR input	
Target	CB	PB	CB	PB	CB	PB	CB	PB
C_5	7.75	37.99	44.88	2.26	1.67	1.56	9.72	7.00
C_6	9.25	5.16	22.31	4.57	4.32	2.42	0.80	18.25
C_7	0.81	0.18	5.82	6.93	1.64	1.16	1.99	0.10
C_8	1.24	0.22	2.99	0.31	0.37	0.52	0.33	0.16

Table 16: Two and three dimensions: absolute relative errors (%) of held-out coefficients at the baseline b_{L}. GFF only: the series resummed directly; with SR input: the curvature ($d = 2$) and slope ($d = 3$) constructions. PB shapes as selected by the rule of Sec. 4.1.

the explicit admissibility diagnostics. They do not validate an instanton amplitude or establish an advantage of supplying the action.

7.2 Order and Borel–Leroy stability

Figure 5 shows the order and b_{L} dependence in $d = 1$. At $K = 8$ the CB curves at orders 6 and 7 differ from CB8 by at most 8.8×10^{-4} and 2.2×10^{-4}; changing b_{L} to 2 or 4 moves the curve by at most 1.4×10^{-4}, while $b_{\mathrm{L}} = 0$ moves it by up to 2.9×10^{-3} (at $K = 4$: 1.8×10^{-2}). The short-range input thus also stabilizes the Borel–Leroy dependence. Varying $1/A_\epsilon$ by $\pm 20\%$ moves the $K = 8$ baseline by at most 1.0×10^{-3} (scale 0.8), and by at most 1.6×10^{-4} for the other variations. In $d = 2$ and 3, adding C_8 changed the curves by at most 8.3×10^{-4} and 1.1×10^{-5} (Sec. 5.3). Small order-to-order changes are stability observations, not an estimate of the total error.

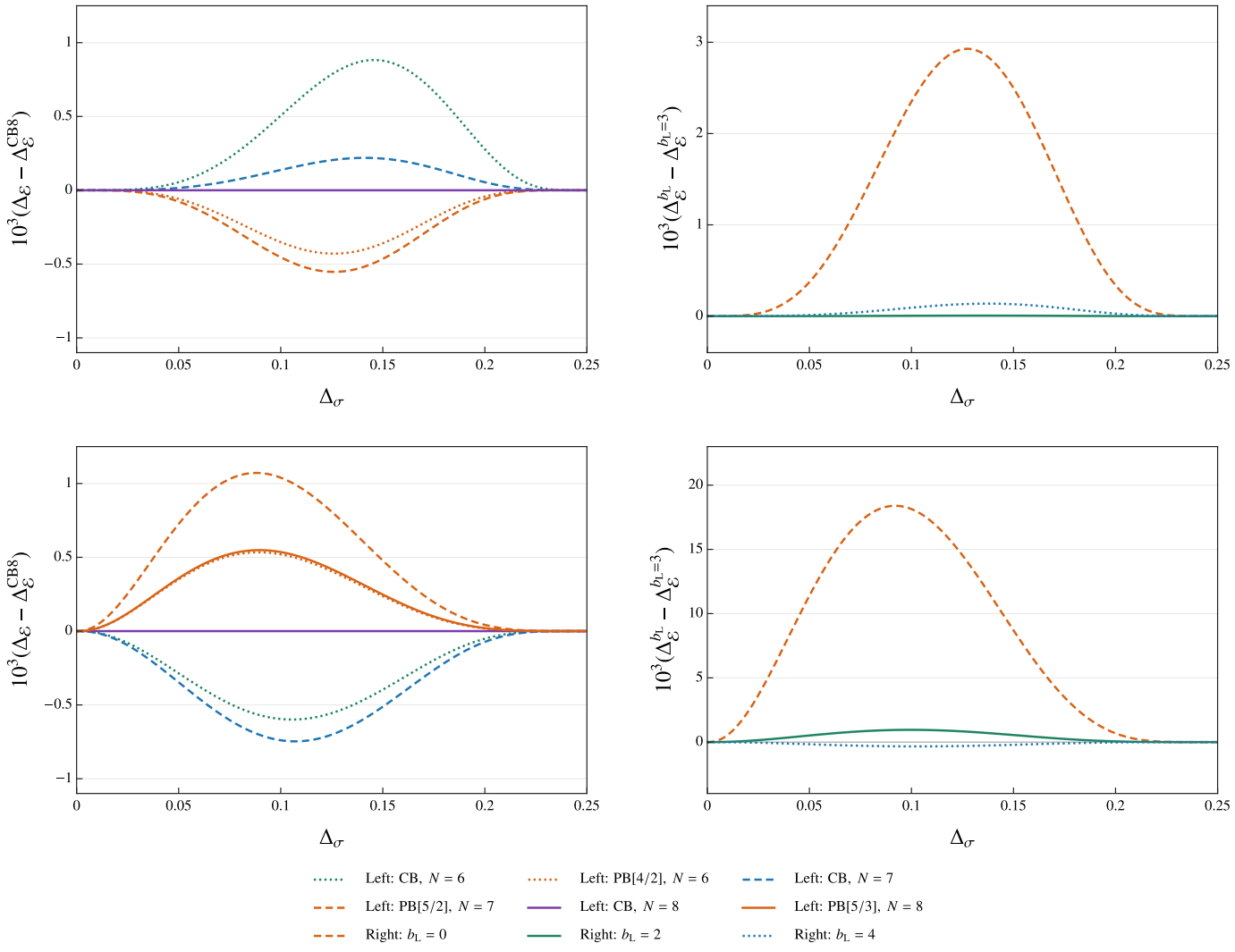


Figure 5: One dimension: finite-order stability. Top row: $K = 8$ baseline; bottom row: the archived $K = 4$ construction (with the final $C_8(1)$). Left: deviations $10^3(\Delta_{\mathcal{E}} - \Delta_{\mathcal{E}}^{\mathrm{CB8}})$ of CB at orders 6, 7 and of the PB controls [4/2], [5/2], [5/3] ($b_{\mathrm{L}} = 3$, drawn only where admissible; at $K = 8$ and order 8 no PB is drawn). Right: deviations of CB8 at $b_{\mathrm{L}} = 0, 2, 4$ from $b_{\mathrm{L}} = 3$. Vertical ranges differ between the right panels.

7.3 Coefficient precision, the final $C_8(1)$ and a ninth coefficient

For fixed CB parameters the reconstruction is affine in the input coefficients, so coefficient sensitivity can be computed exactly from unit responses $J_n = \partial F/\partial C_n$. We report $U = \sum_{n=6}^{8} |J_n|\, \delta C_n$, the sum of absolute responses to the allowances. This would be a worst-case bound only if the allowances were rigorous simultaneous bounds, which they are not; with a covariance matrix Σ the variance would be $J^{\mathsf{T}}\Sigma J$. Four sources of uncertainty must be kept separate: numerical integration (negligible), input coefficients (U), prescription and endpoint construction (the bands), and the theoretical assumptions about the endpoint and the instanton.

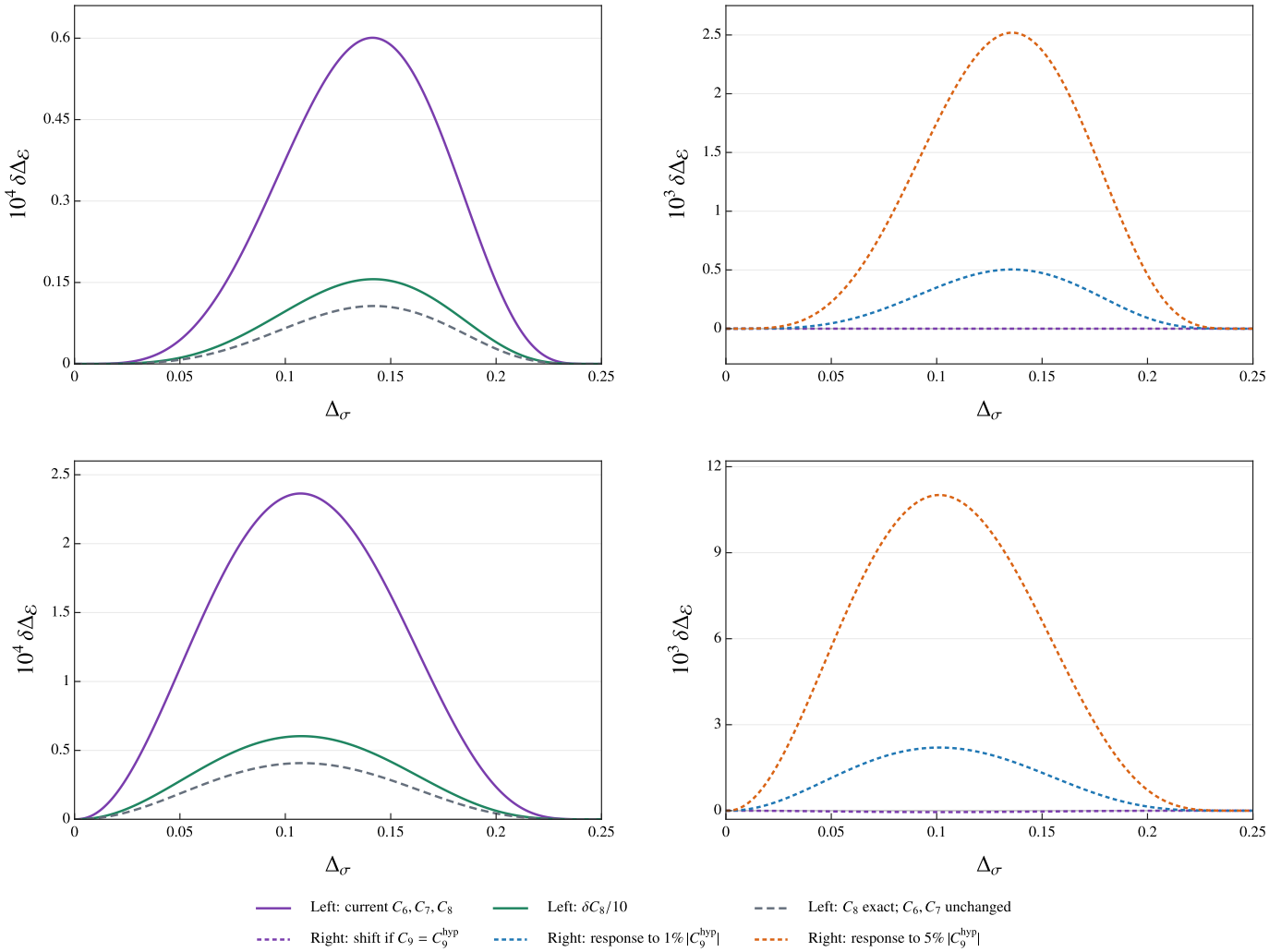


Figure 6: One dimension: coefficient precision. Top row $K = 8$, bottom row $K = 4$ (archived construction, current inputs). Left: $10^4\, U$ for the current allowances ($\delta C_8 = 17000$, $\delta C_7 = 41$, $\delta C_6 = 4.9 \times 10^{-7}$), for $\delta C_8/10$, and with C_8 exact. Right: $10^3\, \delta\Delta_{\mathcal{E}}$ for a hypothetical C_9: the shift if $C_9 = C_9^{\rm hyp}$ and the responses to 1% and 5% of $|C_9^{\rm hyp}|$. Top: $C_9^{\rm hyp} = -2.178401719 \times 10^8$, the CB prediction of the $K = 8$ baseline, so its shift vanishes identically. Bottom: $C_9^{\rm hyp} = -2.190506633 \times 10^8$, the archived $K = 4$ PB[5/3] prediction made with the previous $C_8(1)$, kept as a fixed scenario value. These are sensitivity scenarios, not confidence intervals; C_9 has not been calculated.

At $\Delta_\sigma = 0.1$ and $K = 8$, $U = 3.76 \times 10^{-5}$ with the final allowance; dividing δC_8 by ten gives 9.69×10^{-6}, and the floor with C_8 exact (set by δC_7) is 6.59×10^{-6}. The corresponding values at $K = 4$ are 2.34×10^{-4}, 5.96×10^{-5} and 4.03×10^{-5}. A tenfold better C_8 would thus improve the input budget by a factor 3.9 but leave the prescription envelope, about a hundred times larger, unchanged.

What the final $C_8(1)$ changed. Replacing 8066723.343 ± 33000 by 8059766.723 ± 17000 (a shift of -6957) moves the $K = 8$ baseline down by at most 2.0×10^{-5} (at $\Delta_\sigma = 0.141$; -1.27×10^{-5} at $\Delta_\sigma = 0.1$). It lowers the upper edge of the envelope by up to 1.9×10^{-4}: the width at $\Delta_\sigma = 0.1$ goes from 3.84×10^{-3} to 3.64×10^{-3}, and the maximum width 4.67×10^{-3} is unchanged. It roughly halves the C_8 term of U ($6.02 \times 10^{-5} \to 3.10 \times 10^{-5}$ at $\Delta_\sigma = 0.1$, so that U falls from 6.68×10^{-5} to 3.76×10^{-5}), and it changes the mismatch ratio at $n = 8$ from 0.1578 to 0.1576. With the previous value every program reproduces the archived outputs to 8.4×10^{-15}.

A ninth coefficient. C_9 has not been computed. At fixed CB parameters the ninth-order curve obeys the exact identity

$$F_9(x) - F_8(x) = J_9(x)\,[C_9 - C_9^{\mathrm{CB8}}], \qquad J_9 = \frac{\partial F_9}{\partial C_9}, \tag{7.1}$$

where C_9^{CB8} is the coefficient implied by the eighth-order approximant; supplying that value adds no information. The $K = 8$ baseline implies $C_9^{\mathrm{CB8}} = -2.178401719 \times 10^8$; the only admissible eighth-order PB at $K = 8$, [3/5], predicts -2.181155×10^8, and the $K = 4$ CB8 implies -2.19002×10^8. These are correlated predictions, not a calculation. At $\Delta_\sigma = 0.1$ a change of 1% of $|C_9|$ would move the $K = 8$ curve by 3.5×10^{-4} (at $K = 4$: 2.2×10^{-3}). Adding the ninth order also amplifies the inherited sensitivity to C_8: at $\Delta_\sigma = 0.1$ the C_8 term of U grows from 3.1×10^{-5} to 2.9×10^{-4} at $K = 8$ (from 1.9×10^{-4} to 1.8×10^{-3} at $K = 4$). Simply appending C_9 is therefore not guaranteed to improve the curve: a useful ninth order needs an independently computed C_9, sufficient precision on C_8, and renewed prescription checks.

8 Limitations and open problems

1. **Reporting prescription.** The number $K = 8$ of imposed short-range coefficients, and the range $K = 5$–8 of the envelope, were fixed after the comparison data had been seen. $K = 8$ is motivated a priori only (all short-range coefficients); choosing the narrowest core band would give $K = 7$. Whether the endpoint variable (v or ϵ) belongs in the envelope is open.

2. **Endpoint assumptions.** The constructions assume that the resummed remainder is regular at the short-range end. Logarithms beyond the imposed terms are not excluded. The two-dimensional c_2 carries a formal analytic-MS qualification, the three-dimensional c_1 a conditional spectral range; the one-dimensional C_n^{SR} beyond the published orders are project results with their own assumptions, and the short form of C_8^{SR} is unproved. In $d = 2$ the endpoint construction is the largest uncertainty in mid-window.

3. **The amplitude.** The cancellation of the regulator dependence of the bubble action in the one-loop amplitude (Sec. 3.4) should be derived explicitly. The derivation must keep the integration limits, determinant, measure, legs and counterterms through the simple-pole residue, for the size coordinate and for its rescaled form.

4. **Next order and approach to large order.** The coefficient d_1 needs the two-loop fluctuations, the subleading size integral and the direct mass insertion, and possibly logarithms. Its known part already gives -30% at $n = 8$; further corrections may be needed before eighth order can test K_I.

5. **Dominance.** The Sobolev bound identifies the least-action real saddle; it does not settle complex saddles, integration cycles, multi-instanton sectors, non-instanton factorial mechanisms or the uniformity of the renormalized expansion.

6. **Two and three dimensions.** Only the actions and the power counting are available. A renormalized determinant and amplitude would be needed for amplitude-based tests.

7. **Coefficients.** A covariance model for C_6, C_7, C_8 is missing. A ninth coefficient would be a useful test of the predictions only with adequate precision on C_8 (Sec. 7).
8. **Comparisons.** In $d = 1$ the final HT and NRG values differ at $\Delta_\sigma = 0.1$ and 0.125 by 5.8 and 4.6 times the sum of their quoted errors. The gap is of the size of the dependence of HT on the artifact correction, which the HT error does not include (Sec. 6.2), and its origin is open. HT and NRG resolve the $K = 4$–$K = 8$ difference at several points but lie between the two curves there (Table 13), so they cannot justify the choice of K. In $d = 2$ and 3 HT is available only at finite cutoff; the $d = 2$ mid-window excess of HT over every endpoint construction (about 3×10^{-3} beyond their core bands at $s = 1.6$–1.65) is unresolved, and the cutoff dependence there does not show that it survives at infinite cutoff. Published MC is too imprecise to discriminate, and in $d = 3$ there is a single accepted project MC point. Continuum HT extrapolations and MC with controlled systematics would be the most informative additions.
9. **Interpretation of the $d = 1$ endpoint.** The fixed-point-merger reading of the square-root branch point, and the radius $\simeq \frac{1}{2}$ suggested by the short-range series, should be checked against more coefficients or an independent argument.

A Formulas and parameters

Conformal Borel coefficients. For the resummed series $f(z) = \sum_{j \le m} f_j z^j$ with action A,

$$p_k = \sum_{j=0}^{k} \frac{f_j\,(4A)^j}{\Gamma(j + b_{\mathrm{L}} + 1)}\,(-1)^{k-j} \binom{\alpha - 2j}{k - j}, \qquad \mathcal{R}_m^{\mathrm{CB}}[f](z) = \int_0^\infty e^{-u} u^{b_{\mathrm{L}}} \frac{\sum_{k=0}^m p_k\, w(zu)^k}{[1 - w(zu)]^\alpha}\,\mathrm{d}u, \tag{A.1}$$

which reproduces the Taylor polynomial through order m; for integer $\alpha - 2j$ the generalized binomial coefficient is the polynomial one. The inverse map $w(t) = \sum_{k \ge 1} (-1)^{k-1} \frac{1}{k+1} \binom{2k}{k} (t/4A)^k$ gives an independent round-trip check.

Padé–Borel. With $d_k = f_k / \Gamma(k + b_{\mathrm{L}} + 1)$, the denominator of $[L/M]$ solves $\sum_{j=1}^{M} q_j d_{k-j} = -d_k$ for $k = L + 1, \ldots, m$, and $p_k = \sum_{j=0}^{\min(k,M)} q_j d_{k-j}$ with $q_0 = 1$. The Padé denominator depends nonlinearly on the coefficients, so PB curves are not affine in the inputs.

Series operations and quadrature. Products, powers $(1 - z)^p$ and compositions are formed with truncation after each operation, in 65-digit arithmetic. The Laplace integral uses 240-point generalized Gauss–Laguerre quadrature for weight $u^{b_{\mathrm{L}}} e^{-u}$ (Jacobi matrix with diagonal $2k + b_{\mathrm{L}} + 1$ and off-diagonal $\sqrt{k(k + b_{\mathrm{L}})}$), checked against 120 nodes, adaptive integration and selected 65-digit integrals. For the one-dimensional CB curves the 240-node and adaptive results agree to 1.1×10^{-15}, the 65-digit checks to 5.2×10^{-16}, and the displayed PB curves to 4.0×10^{-15}. Formal re-expansion reproduces the input coefficients. These checks validate the evaluation of each approximant, not its distance from the exact theory.

Grids. The one-dimensional dense grid has 1004 points (1001 equally spaced on $[0, \frac{1}{4}]$ plus the special comparison abscissae); the matched CB/PB comparison uses 502 points. The two- and three-dimensional curves have 754 and 763 stored samples.

Scan parameters. Table 17 collects all prescription parameters.

	$d = 1$	$d = 2$	$d = 3$
Resummed variable	$v = 1 - 2\sqrt{\Delta_\sigma}$	ϵ	ϵ
CB action	$A_v = A_\epsilon/2 = 0.25703245$	$A_\epsilon = 9/8$	$A_\epsilon = 1.92343326$
Endpoint input	$C_1^{\mathrm{SR}}, \ldots, C_8^{\mathrm{SR}}$ $(K = 8)$	1, 0, c_2	E, c_1
Endpoint controls	$K = 4$–7, endpoint-ϵ	tangent, value	value
Without SR input	GFF-only	GFF-only	GFF-only
Orders	$6, 7, 8$	$6, 7, 8$	$6, 7, 8$
Core b_{L}	$2, 3, 4$	$2.5, 3.5, 4.5$	$3, 4, 5$
Wide b_{L}	$0, 1, \ldots, 6$	$0, 1.5, 2.5, \ldots, 6.5$	$0, 2, 3, \ldots, 7$
α (core; wide)	$-1, 0, 1$; $-2, \ldots, 2$	$-1, 0, 1$; $-2, \ldots, 2$	$-1, 0, 1$; $-2, \ldots, 2$
Action variations	$1/A \times 0.8, 0.9, 1.1, 1.2$	same	same
PB control	[5/3] $(K \leq 7)$	PB8 [5/3]	PB7 [3/4]; PB8 [5/3] at $b_{\mathrm{L}} = 0$
Stored grid	1004 points	754 points	763 points

Table 17: Resummation parameters. The baseline uses order 8, the middle core b_{L} and $\alpha = 0$; the PB controls use the baseline b_{L} unless stated.

One-dimensional comparison data. Table 18 lists every one-dimensional value compared in this note, with its residual from the baseline.

Table 18: One-dimensional comparison data [4, 5, 6, 21, 22], labels as in Table 12. "err": for HT the half-spread of the accepted fits, for NRG the chain-depth, kept-state and discretization errors added linearly, for MC σ_{sys}, for published MC the quoted error; "stat": the MC statistical error. A dot means none supplied. MC Δ_{eff}: finite-size exponents of the largest lattice pair (sizes in parentheses), not infinite-size values. The last column is $10^3(\Delta_{\mathcal{E}} - \Delta_{\mathcal{E}}^{\mathrm{baseline}})$.

Method	Δ_σ	$\Delta_{\mathcal{E}}$	err	stat	status	resid.
HT	0.249	0.4993436	9.3×10^{-8}	·	final	−0.00
HT	0.24	0.4942198	2.6×10^{-6}	·	final	+0.00
HT	0.225	0.4880804	5.2×10^{-6}	·	final	−0.00
HT	0.2	0.4830244	4.7×10^{-6}	·	final	−0.09
HT	0.175	0.4839086	1.5×10^{-5}	·	final	−0.33
HT	0.15	0.4912086	3.3×10^{-5}	·	final	−0.50
HT	0.125	0.5061186	4.3×10^{-5}	·	final	−0.31
HT	0.1	0.5304771	1.2×10^{-4}	·	final	+0.23
HT	0.075	0.5673260	2.0×10^{-4}	·	one-sided	+0.70
HT	0.05	0.6206712	·	·	unstable ($N = 40$)	−1.56
HT	0.025	0.7105771	·	·	no limit ($N = 40$)	−2.86
HT	0.01	0.8091106	·	·	no limit ($N = 40$)	−0.64
HT	0.001	0.9693379	·	·	no limit (not shown)	+31.91
NRG	0.249	0.508030	·	·	not converged	+8.69
NRG	0.24	0.498922	·	·	not converged	+4.70
NRG	0.225	0.489668	·	·	not converged	+1.58
NRG	0.2	0.482980	7.1×10^{-4}	·	final	−0.14
NRG	0.175	0.483900	3.5×10^{-4}	·	final	−0.34
NRG	0.15	0.491106	1.1×10^{-4}	·	final	−0.61
NRG	0.125	0.505672	5.4×10^{-5}	·	final	−0.76
NRG	0.1	0.529608	3.3×10^{-5}	·	final	−0.64
NRG	0.075	0.566260	3.0×10^{-5}	·	final	−0.37
NRG	0.05	0.622115	2.7×10^{-5}	·	final	−0.12
NRG	0.025	0.713422	2.9×10^{-5}	·	final	−0.01
NRG	0.01	0.809747	4.7×10^{-5}	·	final	−0.00
NRG	0.001	0.937424	1.1×10^{-4}	·	final	−0.00
MC	0.0125	0.78404	0.01003	0.00077	cross-check unavailable	−5.11
MC	0.025	0.71115	0.00959	0.00049	accepted (σ_{sys}-dominated)	−2.28
MC	0.0375	0.65904	0.00425	0.00051	cross-check unavailable	−2.42

continued on the next page

Method	Δ_σ	$\Delta_\mathcal{E}$	err	stat	status	resid.
MC	0.05	0.61987	0.00220	0.00055	accepted	−2.36
MC	0.0625	0.59003	0.00150	0.00035	cross-check unavailable	−1.36
MC	0.075	0.56694	0.00139	0.00037	accepted	+0.31
MC	0.0875	0.54761	0.00194	0.00032	accepted	+1.05
MC	0.1	0.53248	0.00242	0.00020	accepted ($\sigma_{\rm sys}$-dominated)	+2.23
MC	0.1125	0.52020	0.00217	0.00025	accepted ($\sigma_{\rm sys}$-dominated)	+3.17
MC	0.125	0.50710	0.00193	0.00035	accepted	+0.67
MC	0.1375	0.49856	0.00180	0.00030	accepted ($\sigma_{\rm sys}$-dominated)	+0.48
MC	0.15	0.49325	0.00306	0.00041	cross-check failed	+1.54
MC	0.1625	0.48761	·	0.00058	conditional	+0.47
MC	0.1727335	0.48368	0.00078	0.00052	accepted	−0.97
MC	0.175	0.48460	0.00067	0.00035	cross-check failed	+0.36
MC	0.1875	0.48263	·	0.00030	conditional	−0.28
MC	0.225	0.47912	·	0.00068	conditional	−8.96
MC $\Delta_{\rm eff}$	0.2125	0.4784	·	0.0015	model-limited (256k→ 512k)	−6.43
MC $\Delta_{\rm eff}$	0.2375	0.4826	·	0.0015	model-limited (128k→ 256k)	−10.41
Uzelac 2001	0.05	0.600	0.010	·	published	−22.23
Angelini 2014	0.0625	0.5876	0.0013	·	published	−3.79
Uzelac 2001	0.1	0.53	0.02	·	published	−0.25
Uzelac 2001	0.15	0.49	0.02	·	published	−1.71
Angelini 2014	0.1727335	0.494	0.014	·	published	+9.35
Uzelac 2001	0.2	0.50	0.02	·	published	+16.88

B Reproduction

Everything in this note is reproduced from the folder `Resummation/` of the public repository. Paths below are relative to it. The programs regenerate the resummations, the sensitivity analyses and the comparisons from the coefficients and the comparison data; they do not regenerate graph integrals, HT spectra, NRG runs or MC samples, which come from the sibling folders `GFF/`, `HT/`, `NRG/` and `MC/` (frozen copies in `data/inputs/`, listed with their sources in `data/inputs/INPUTS.json`).

Packages. `code/resum_d1/`: the one-dimensional resummation of Secs. 4–7 (endpoint construction with the short-range series through $C_K^{\rm SR}$, the $C_8(1)$ switch, curves, envelopes, matched CB/PB comparisons, held-out tests, C_9 scenarios, allowance budgets, reference residuals, the instanton-mismatch table and the $K = 4/K = 8$ table). `code/resum_d23/`: the two- and three-dimensional resummations, the GFF-only CB3 curves and the table of published MC values. `code/instanton/`: the critical bubble, the amplitude K_I and its benchmarks (Sec. 3). `code/comparison/`: the comparison numbers of Sec. 6. The production files ship unchanged with their production layout and a dated sha256 list (`public/SNAPSHOT.sha256`); the tools added for the public replay are in each package's `public/` and `public.mk`.

Commands. From `Resummation/`:

```
make test           # fast tests of every package (1 min)
make replay         # every quoted number, shipped records (1 min)
make replay-full    # rerun the resummations, compare outputs (3 min)
make check-inputs   # frozen inputs against the sibling folders
```

`make replay` recomputes every number of the sections on resummation and on the comparison of the paper [1] and of its paragraph on higher orders, and the entries of its results tables that come from this folder: the resummed values in $d = 1, 2, 3$, the HT values at the largest cutoff in

$d = 2, 3$ and the MC Binder exponents in $d = 2$ (`results/quoted_numbers.json`). It compares each value, rounded half-up to the printed digits, with the printed text (the MC exponents with their errors by the PDG rule). The map from each number to its record, command and tolerance is in `results_map.md` (per package in `code/<package>/public/results_map.md`). It also regenerates, byte for byte, the tables derived from the frozen inputs (the table of one-dimensional comparison data, the $K = 4/K = 8$ table of Sec. 6, the published-MC table and the comparison tables in `results/comparison/`), and checks the data behind the two figures of the paper (`results/figures/*.csv`). `make replay-full` reruns the one-dimensional chain (`code/resum_d1/update_2026-10-13/C8_1_ext/run_chain.sh`, about 2 min on one core) and the ten 2d/3d scripts; every output must agree with the shipped one to $10^{-12} + 10^{-12}|x|$ (fits of the coefficient-model analysis to 10^{-6} relative), and the quoted numbers are recomputed from the rerun. Measured on the symmetry cluster with Python 3.12, numpy 2.3, scipy 1.17: all differences below 5×10^{-15} except those fits (2×10^{-6}).

Numbers of this note that the paper does not quote. They are read from the records of the packages: the one-dimensional curves, bands and key points from `code/resum_d1/update_2026-10-13/C8_1_ext/pipeline_K8/results/` (`curve_K.csv`, `key_points_K.csv`, `ablation_K.csv`, `held_out_K.csv`, `c9_*`, `precision_budget_K.csv`, `comparison_points_K.csv`) and `.../instanton_mismatch/`; the two- and three-dimensional ones from `code/resum_d23/\{2d,3d\}/results/`; the amplitude from `code/instanton/results.json`. `make replay-full` regenerates all of these records.

Figures. The six figures are rendered with Wolfram Language and MaTeX (`vendor/MaTeX/`) by `code/resum_d1/plot_lri_comparisons.wl` (Figs. 1, 4, 5 and 6), `code/resum_d1/plot_lri_dispersion.wl` (Fig. 3) and `code/resum_d23/plot_lri_2d3d.wl` (Fig. 2), through `run_plot.wls`; `make figures` rebuilds them where `wolframscript` and TeX are available. The replay checks the plotted data, not the pixels.

Superseded constructions. The archived $K = 4$ construction (short-range series through C_4^{SR}) and the previous $C_8(1) = (8.067 \pm 0.033) \times 10^6$ appear only where this note compares with them; the previous-$C_8(1)$ outputs are not shipped (they are regenerated by `LRI_C8_1=previous`).

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